

GLAUCOMA VISION → Silent killer of vision

POAG → slow, progressive & painless ↓se in vision → **IRREVERSIBLE**

Screening: after 40yrs of age annually screen for Glaucoma.

PSEUDOEXFOLIATION (PEX) syndrome : MCC of 2° OAG

MCC of 2° ACG → DM retinopathy

C/hallmark → deposition of whitish Ab (N) pseudoxfoliative debris on the anterior lens capsule
→ deposition on TM → 2° OAG
" " Zonules → Zonolysis → Lens subluxation

Rx: Medical Mx of POAG.

JIPMER

Q. Which of the following condition has pathognomonic finding in Retina?

a) Sickle cell anaemia

b) Infective Endocarditis

c) Wilson's d/s

d) Polycythemia rubra vera.

SICKLE CELL Retinopathy

1) Sunburst Retinal H'mage

2) Sea fan configuration of neovascularisation.

AIIMS:

Q. Most radioresistant cells in retina?

a) Retinal pigment epithelium

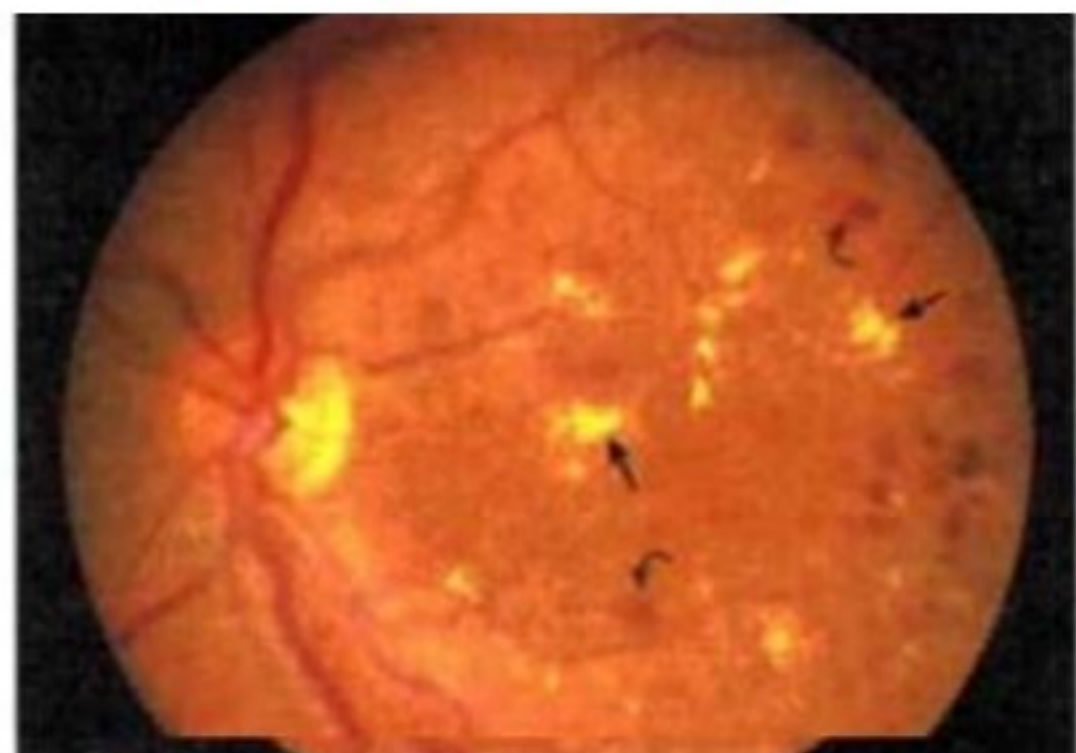
b) Ganglion cell layer

c) Rods & Cones

d) Bipolar cells

sol. most radiosensitive cells in Retina → Retinal vascular endothelial cell
> Rods.

Q. A 58-year-old woman is referred to an ophthalmologist because of vision loss. She describes previous episodes of acute vision impairment that gradually improved over days. An image of her retina is shown below. If this patient were to undergo renal biopsy, what are the pathological findings associated with d/s



Diabetic Retinopathy is often preceded by nephropathy
∴ Dx of diabetic retinopathy with a simple fundoscopic examination can predict the onset of other microvascular complications.

a) Nodular glomerulosclerosis, mesangial expansion, and basement membrane thickening } Diabetic Retinopathy

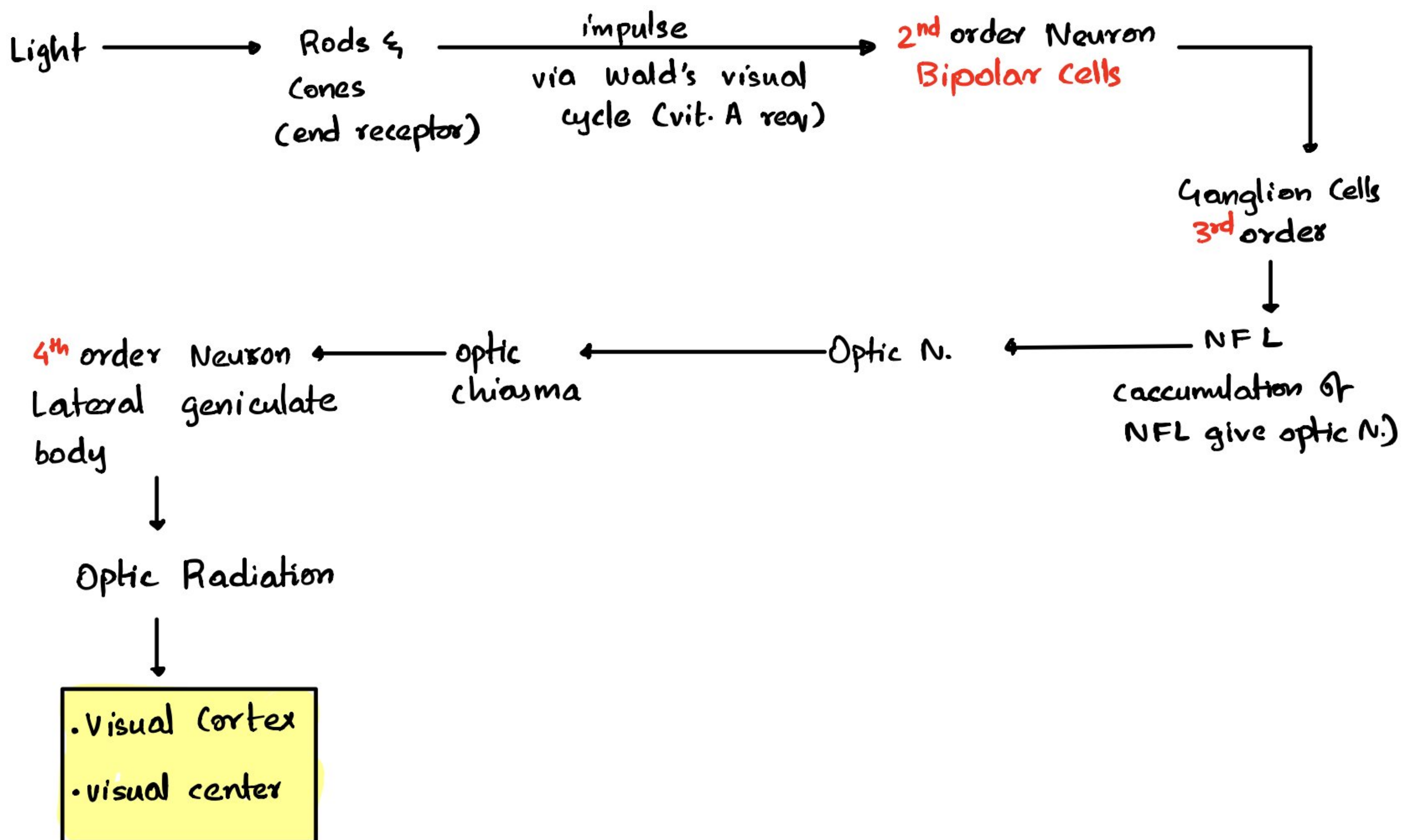
b) scattered crescentic lesions with fibrin and plasma proteins

c) Vascular intimal thickening and hyaline deposition

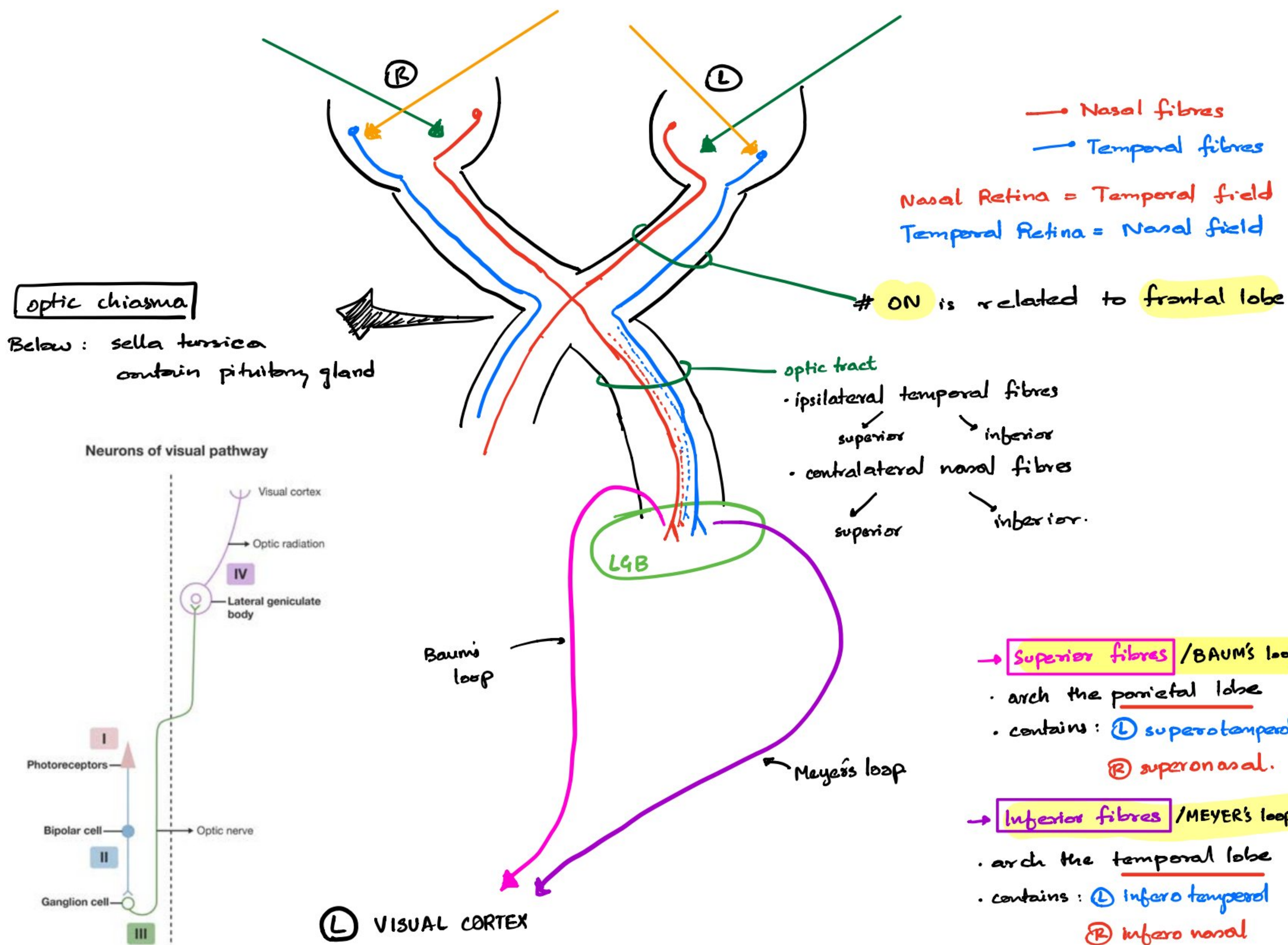
d) Subepithelial immune complex deposits

Visual Pathway

1st Order Neuron



Speed of Visual impulse = 124ms



Visual cortex: → Occipital Lobe

1° v. cortex: Brodman Area # 17

Area V₁

2° v. cortex / Visual association area

Brodman Area # 18 & 19

Area V₂ & V₃

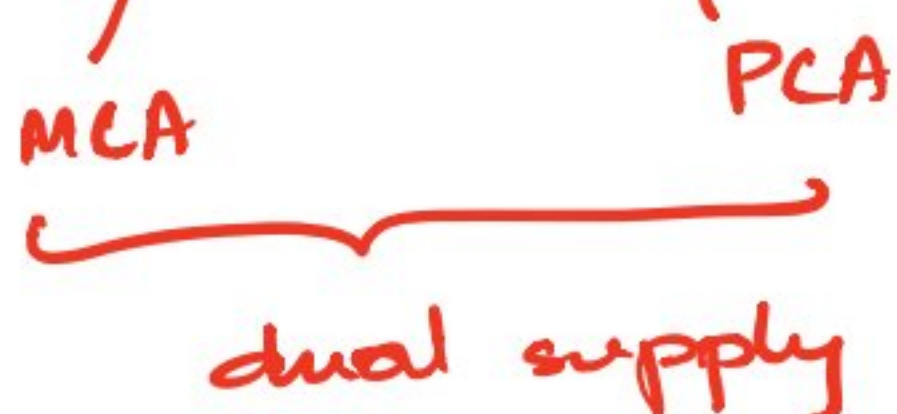
• Area V₄ → Color vision

→ **ACHROMATOPSIA**

Location:

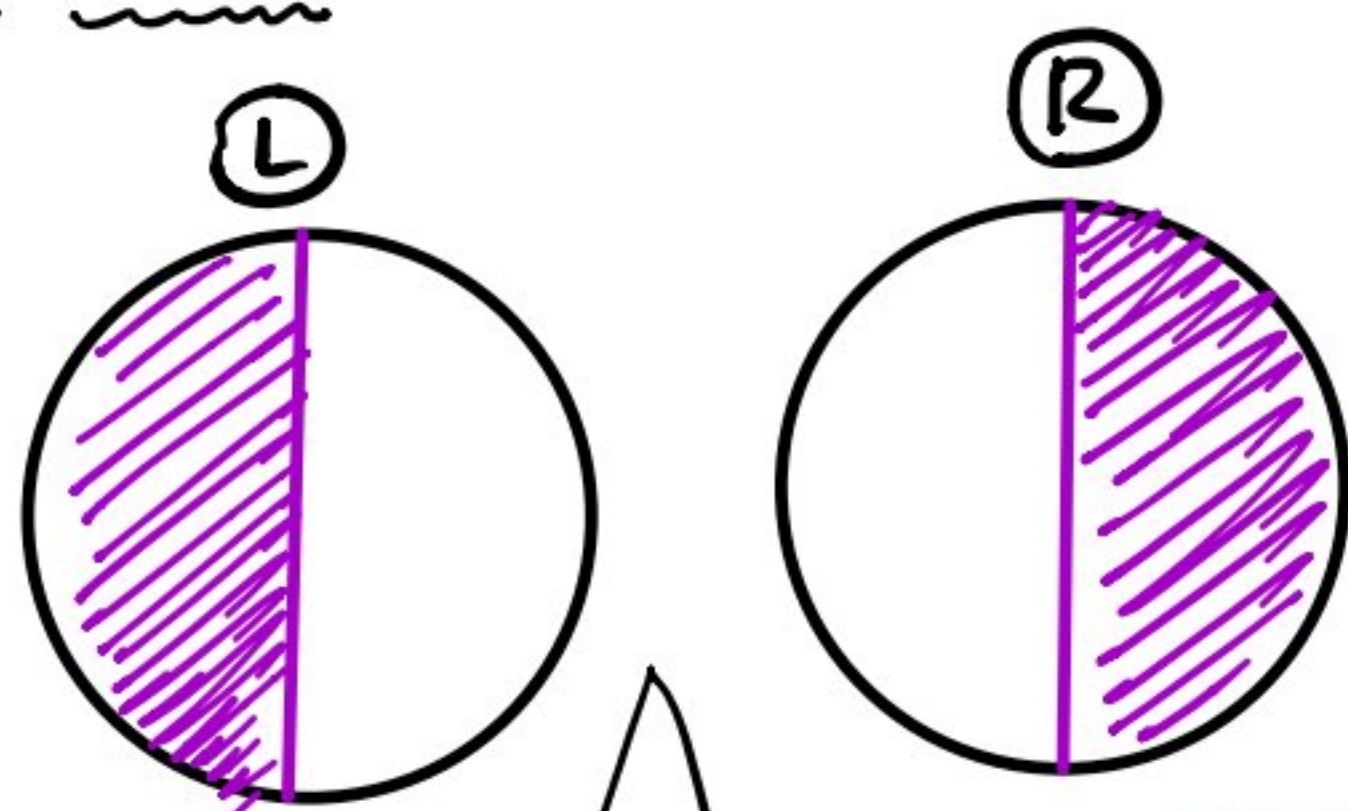


Blood supply:



Lesions of Visual Pathway:

① Optic Chiasma:



BITEMPORAL HEMIANOPIA

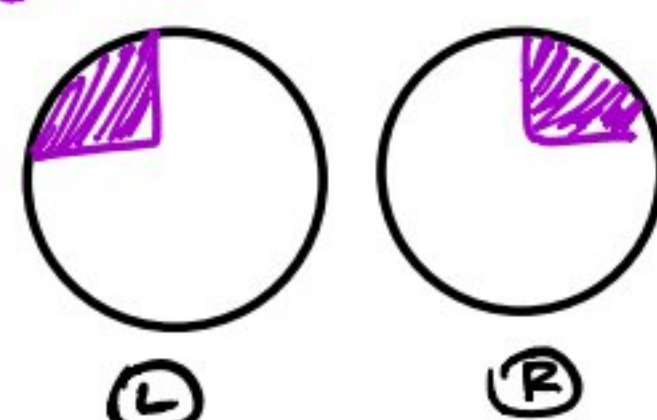
(Heteronymous → always only chiasma)

cause:

- pituitary lesion
- craniopharyngioma

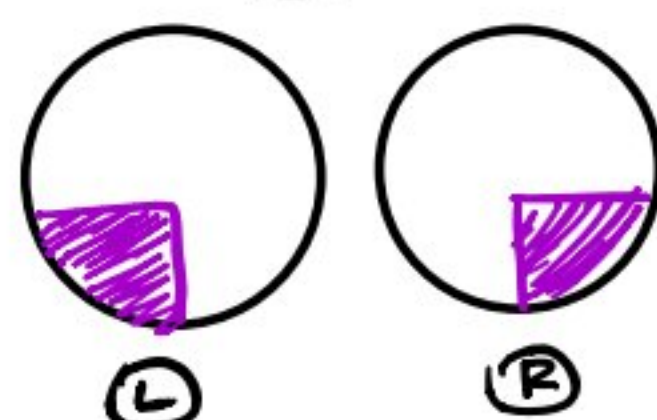


P.T. affects first 1/2 inferior nasal fibres



B/L SUPERTEMPORAL QUADRANOPIA

← early pituitary tumor.

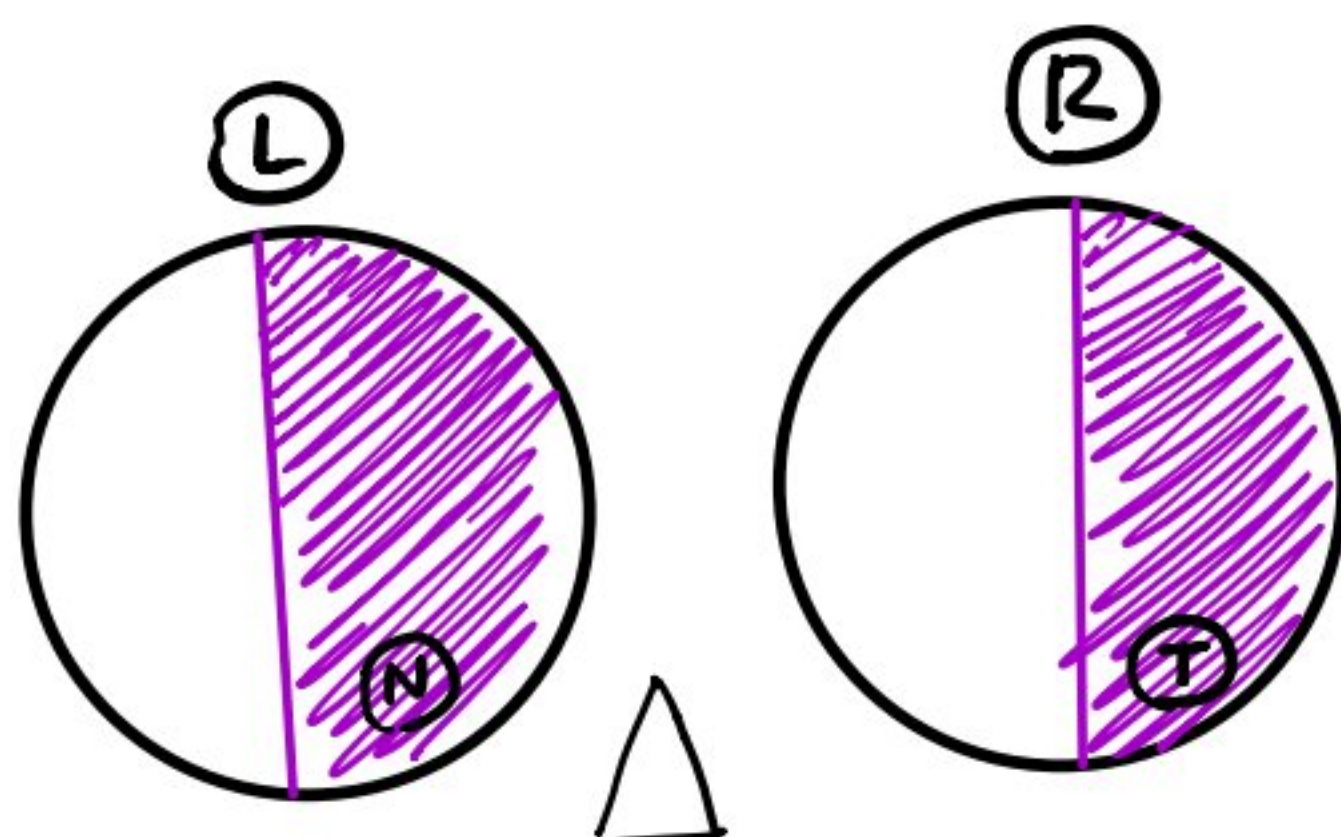


B/L INFEROTEMPORAL QUADRANOPIA

← early craniopharyngioma.

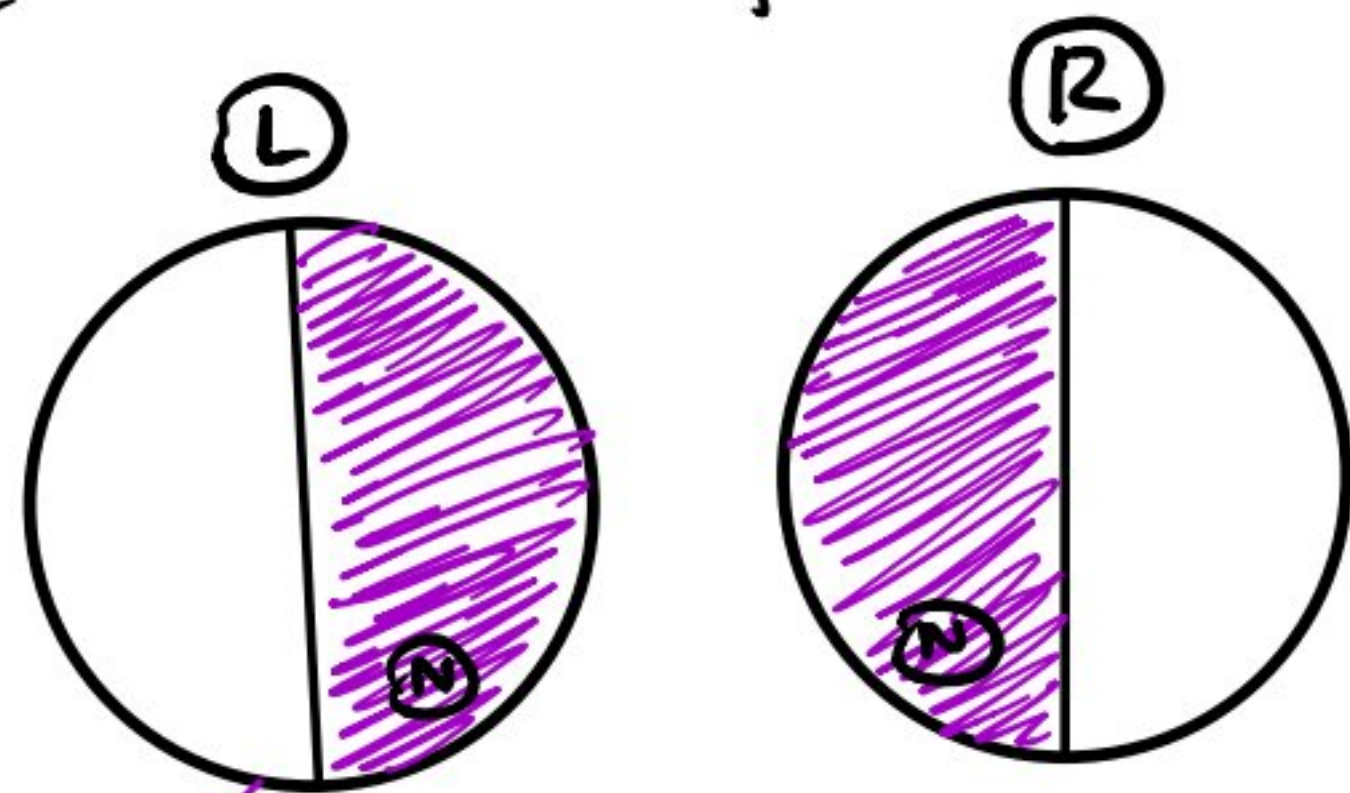
Any lesion posterior to Optic Chiasma gives contralateral visual field defect

② Optic Tract



③ HOMONYMOUS HEMIANOPIA

Q: The most likely site of lesion?



BINASAL HEMIANOPIA

Heteronymous → always chiasma

- Same side = Homonymous
- Extent same = Congruous
- Same side

a) L OT

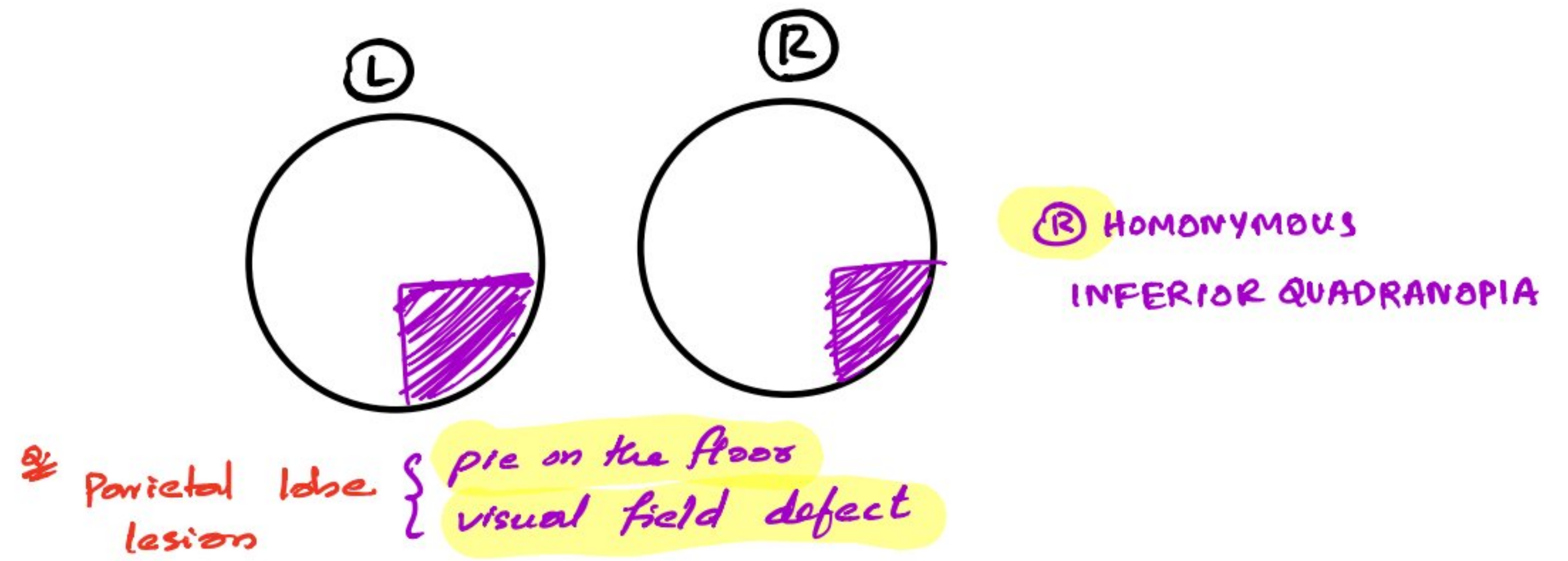
b) R OT

c) Chiasma

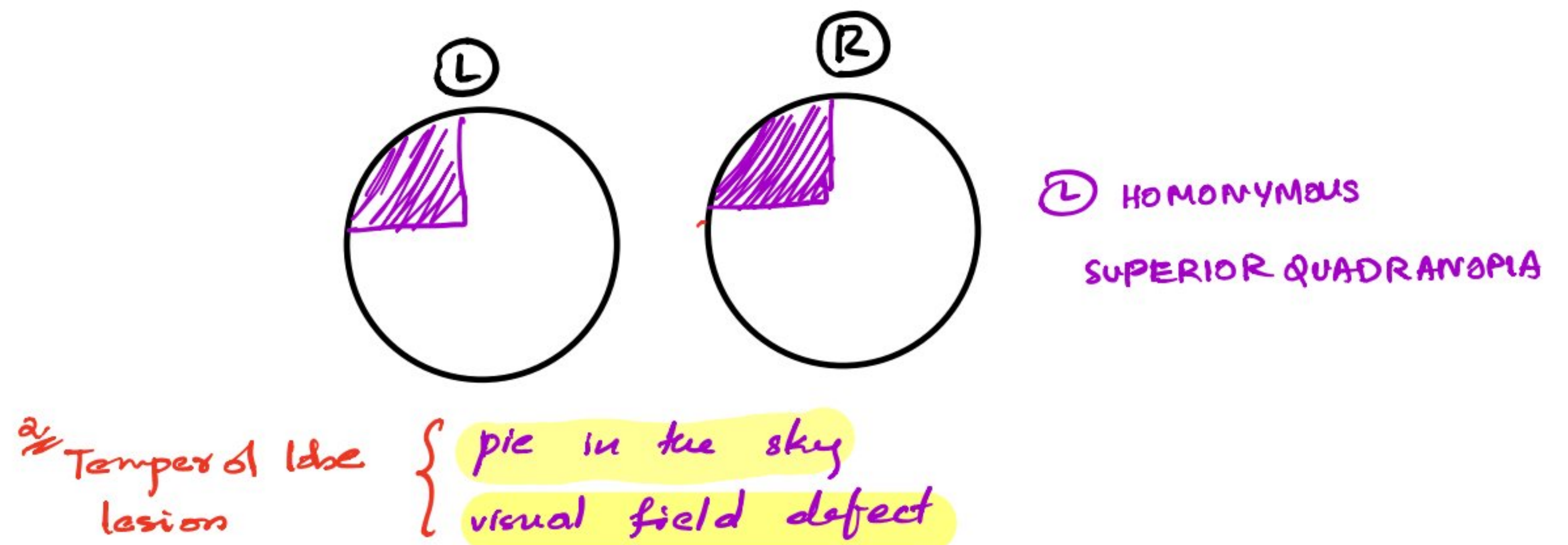
d) None of the above

③ Lesion of optic radiation

eg: Lesion of ① parietal lobe → ① superotemporal fibres → ① inferonasal field
 Baum's loop ② supero nasal fibres → ② infero temporal field.

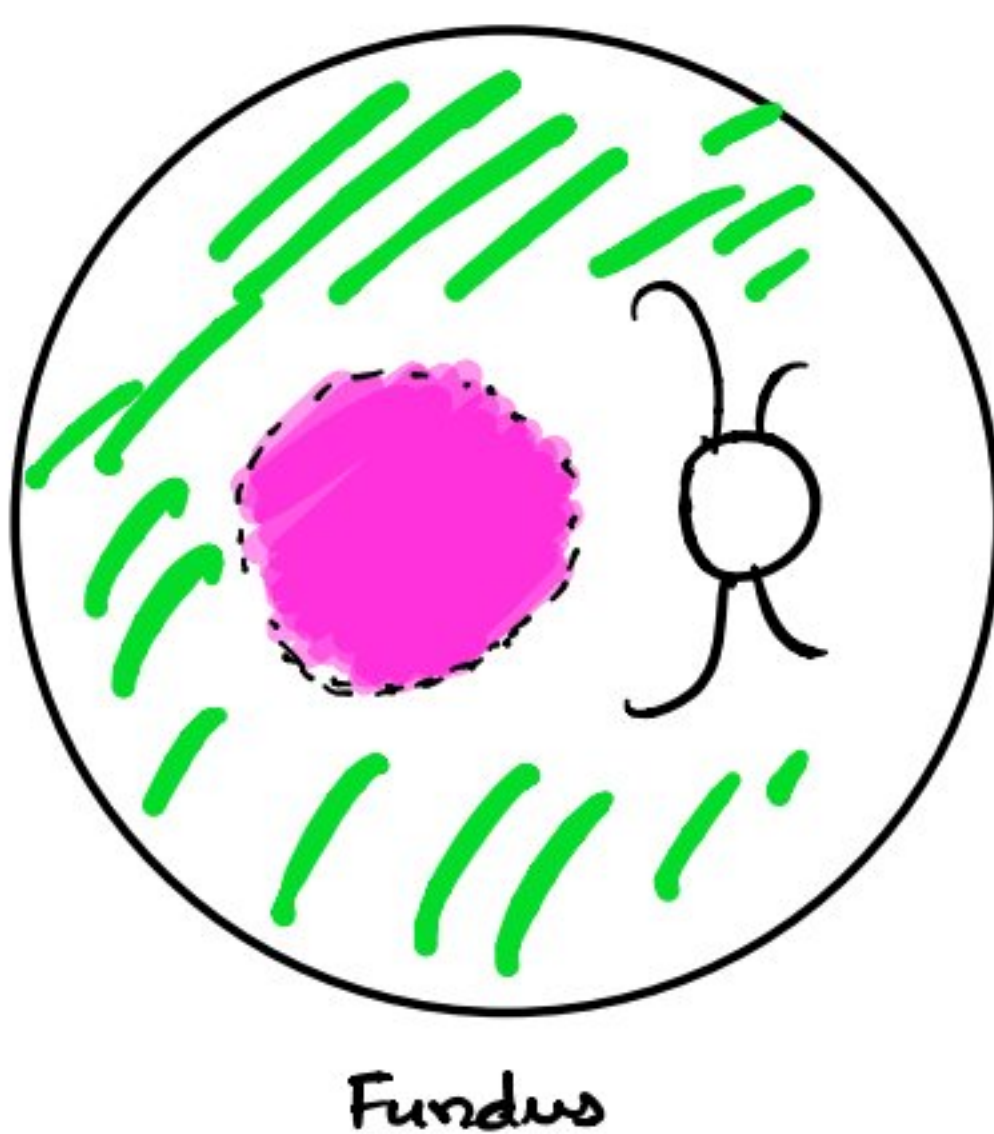
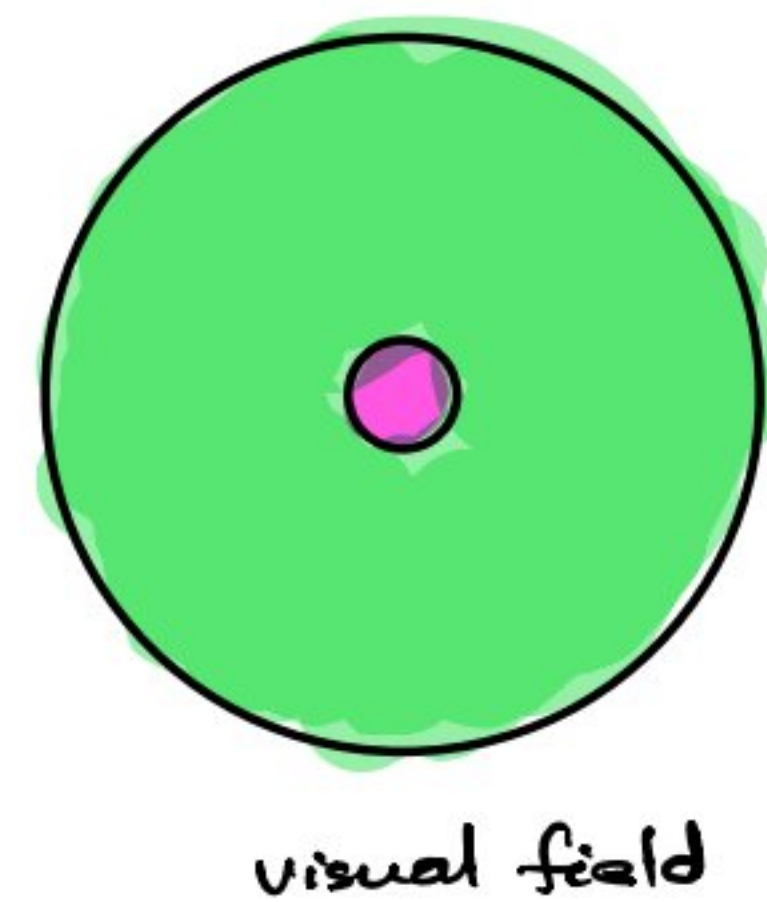
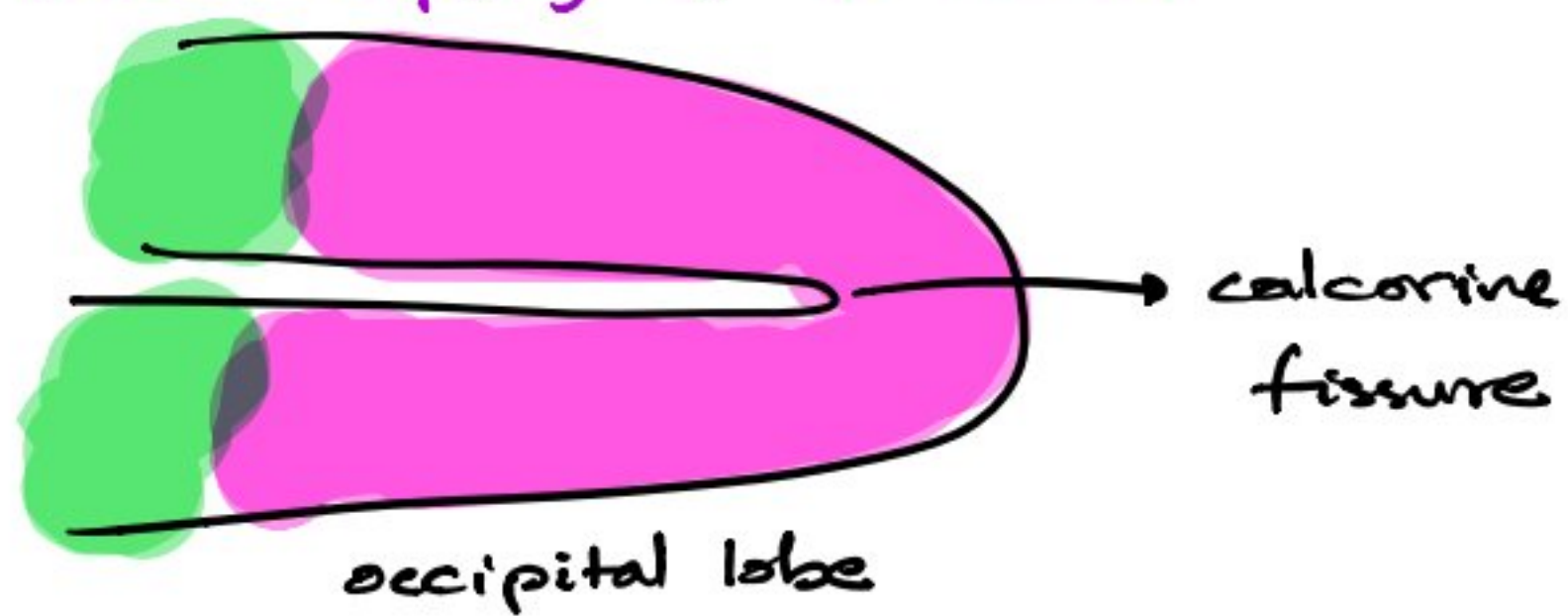


eg: ② Meyer's loop lesion → ② infero temporal fibres → ② superonasal field
 ① infero nasal fibres → ① supero temporal field



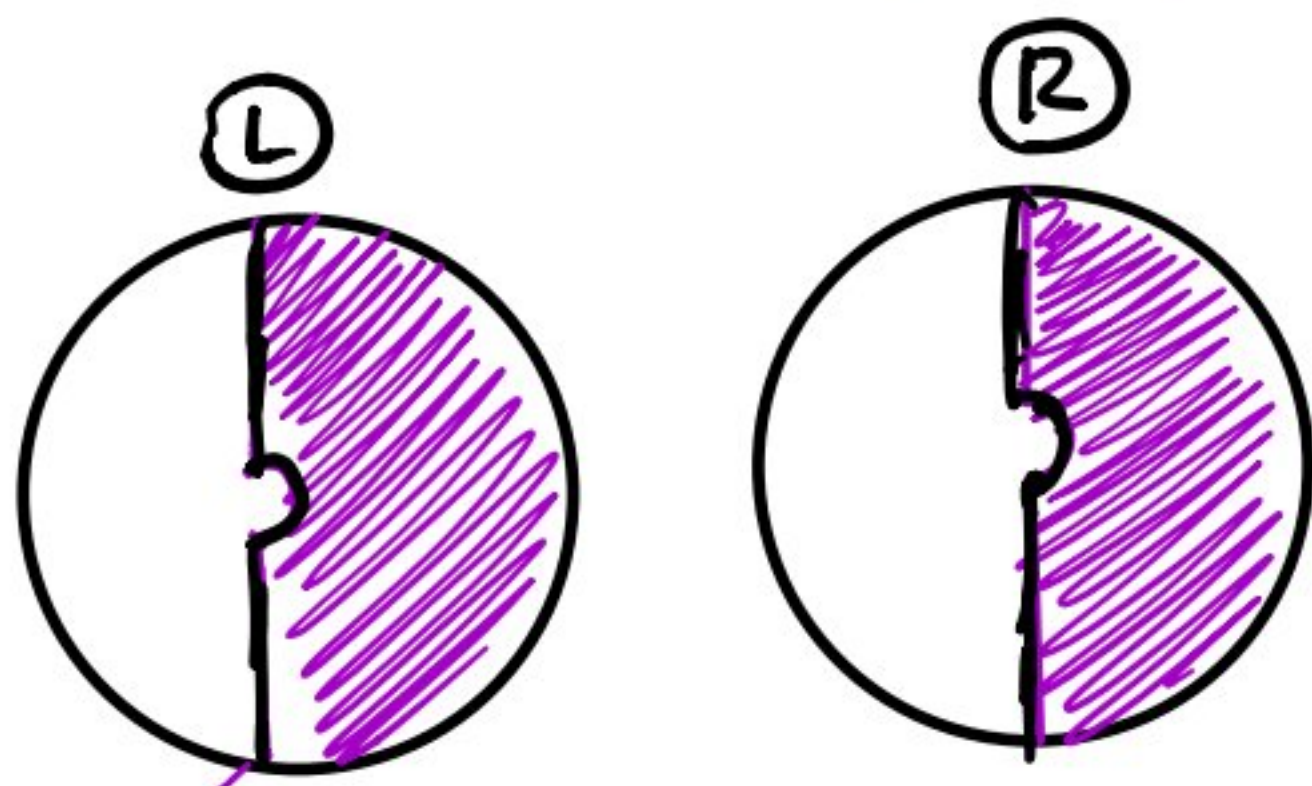
④ Lesion of visual cortex:

- Extensive representation of macula in visual cortex
- Dual Blood supply of V. cortex

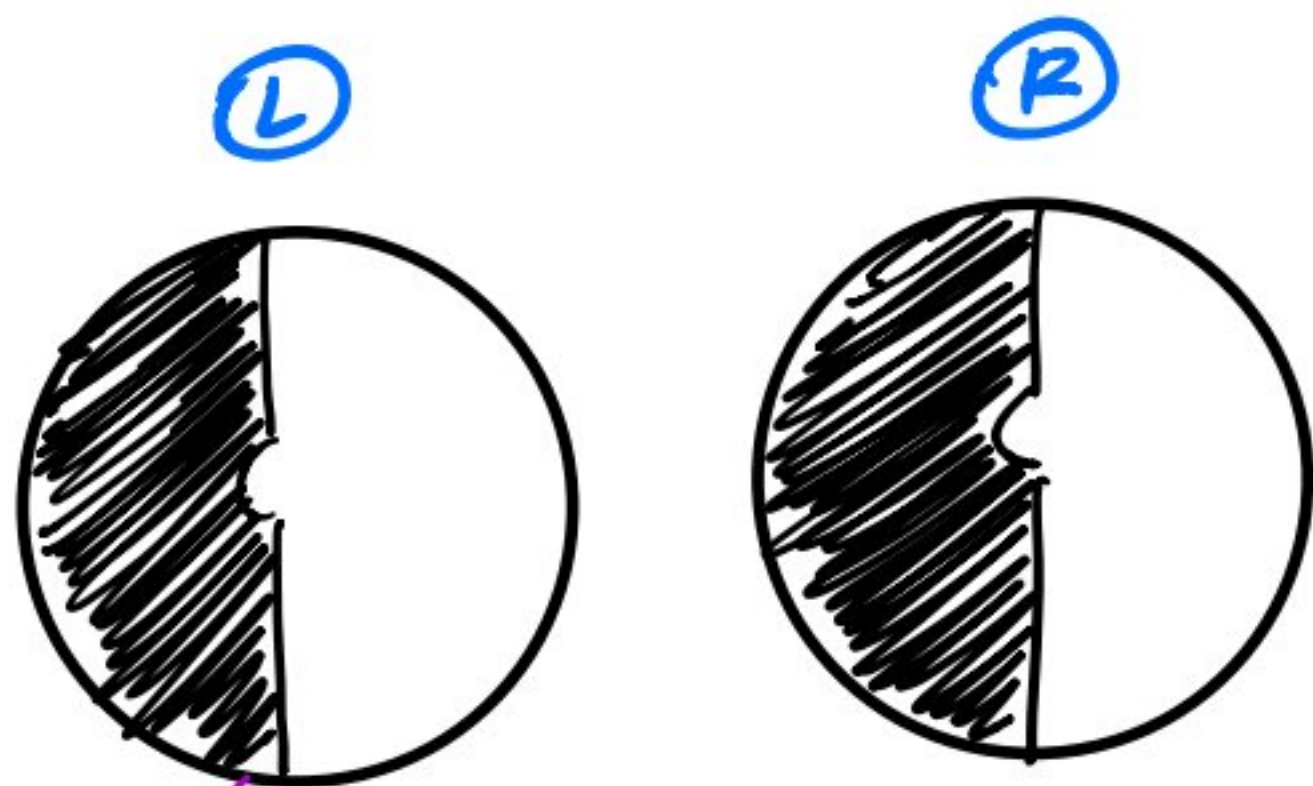


Ex: Lesions of ② visual cortex:

↳ ② homonymous hemianopia
 + macular sparing.

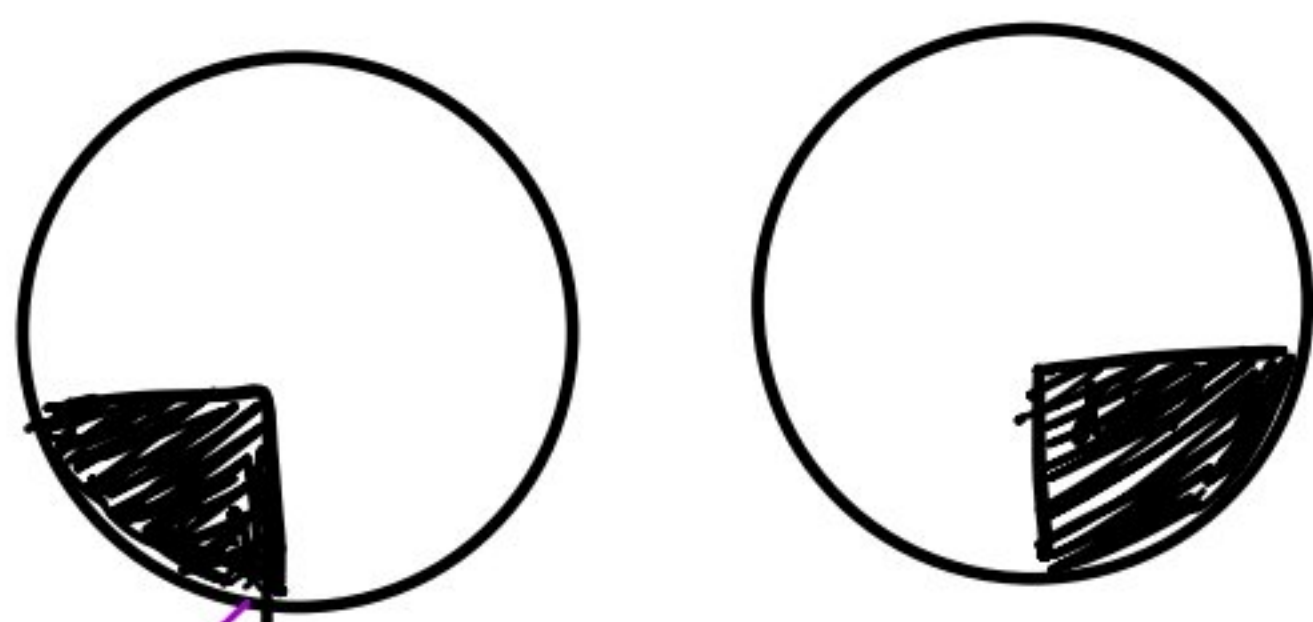


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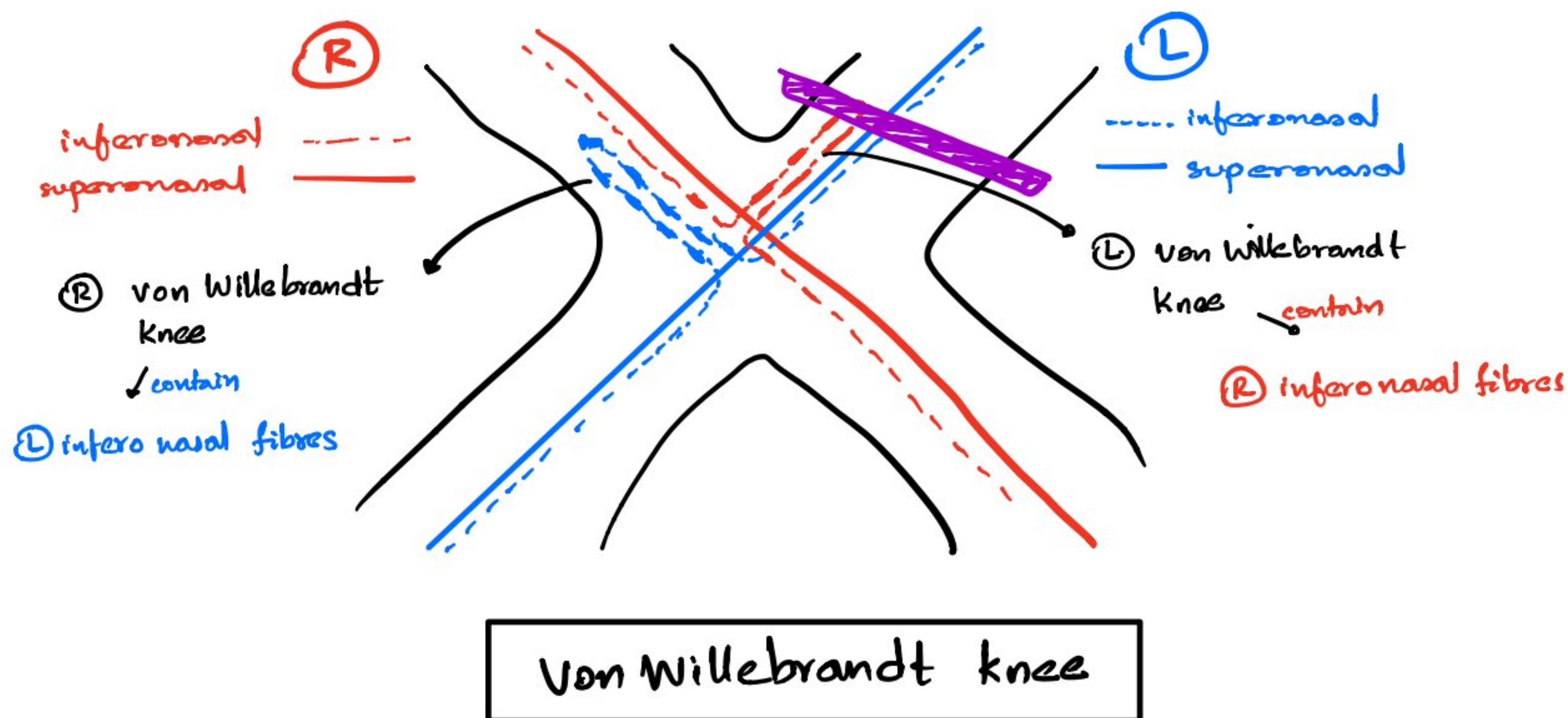


sol. ② visual cortex

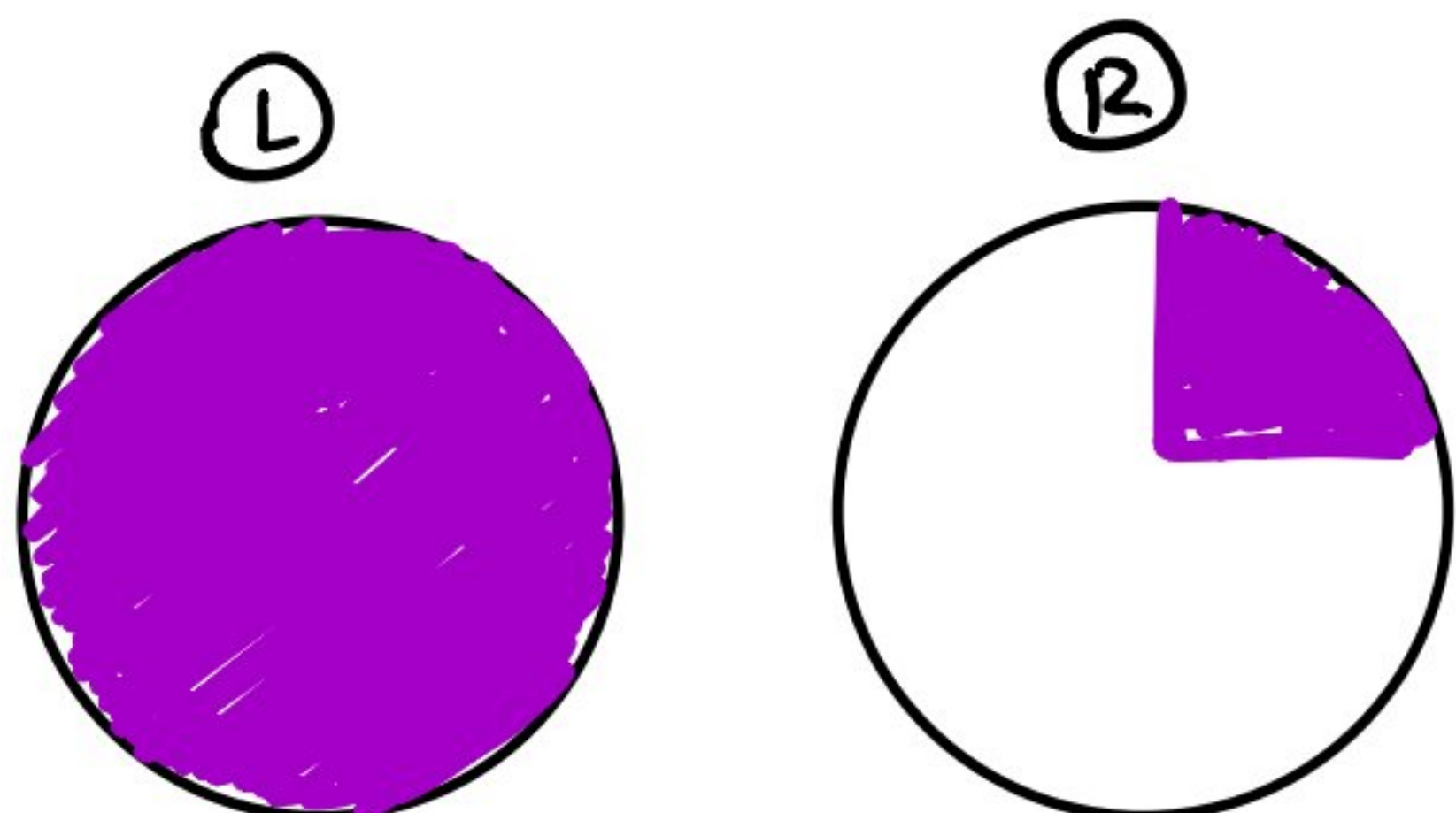
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a) ② Meyer's loop
 b) ② Baum's loop
 c) Craniopharyngioma
 d) Pituitary tract



⑤ Complete ② optic N. lesion.



② complete Anopia

+

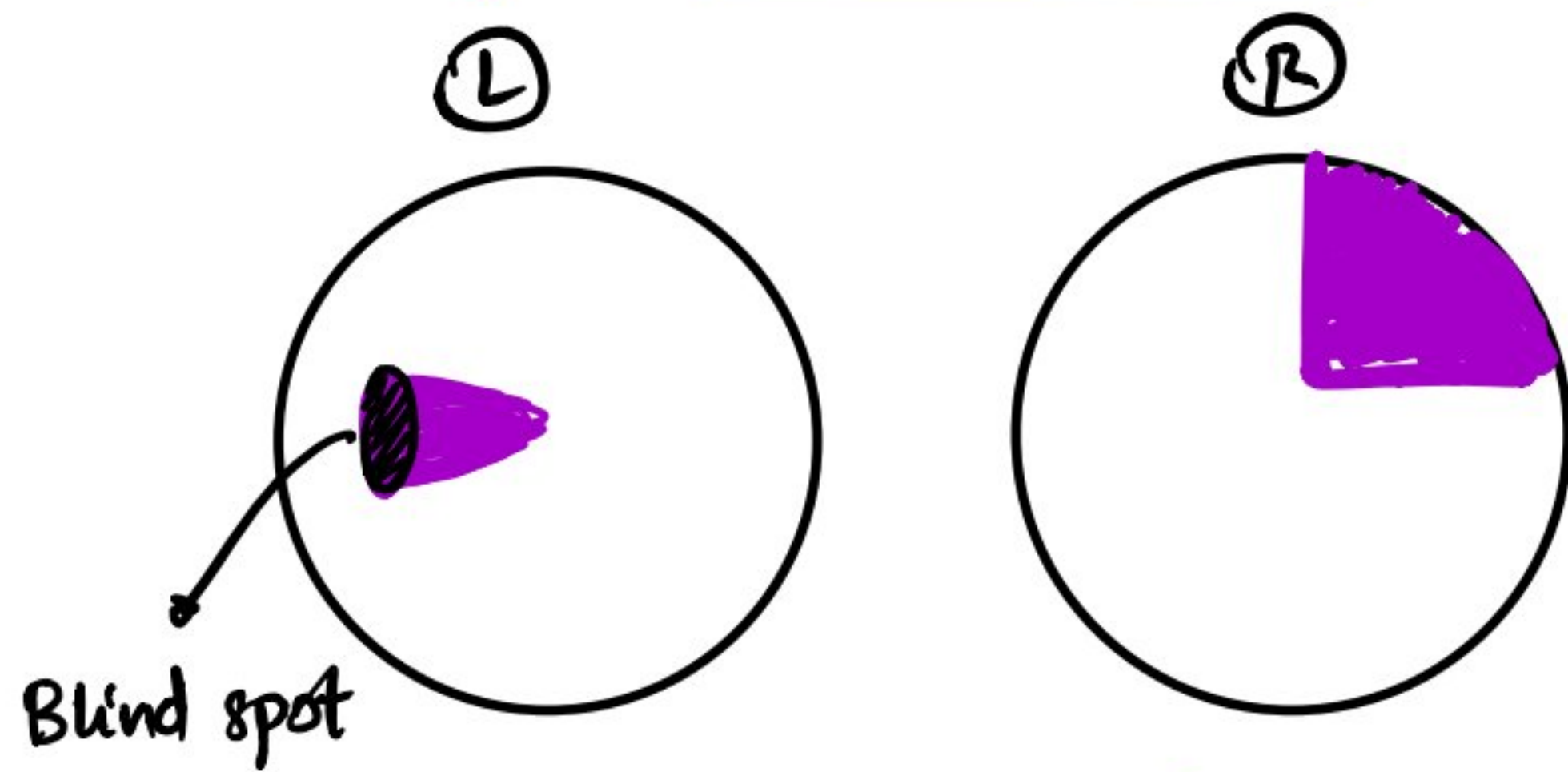
② superotemporal quadrantanopia

JUNCTIONAL SCOTOMA

cause:..

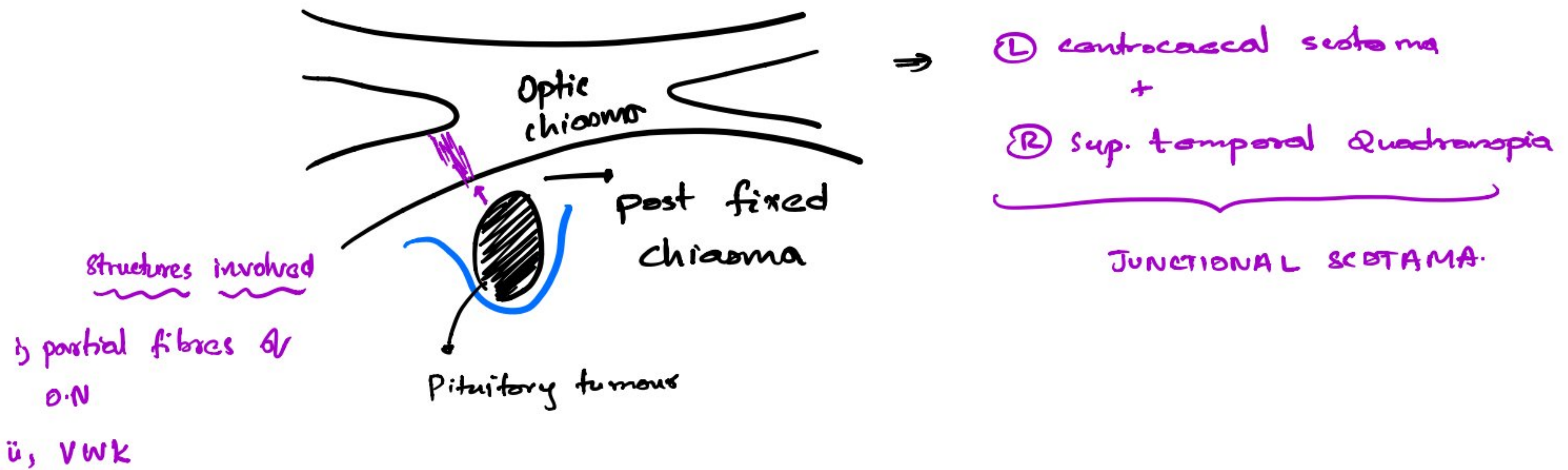
② TUBEROUS MENINGIOMA

① optic neuritis ⇒ doesn't affect all the fibres of optic N.
gives **centrocaecal scotoma**



① centrocaecal scotoma + ② superotemporal quadrantanopia } **Junctional scotoma.**

Post Fixed Chiasma



Colour Vision Pathway: ≈ visual pathway

Differences

- 1) Impulse formed only on cones
- 2) X ganglionic cells are involved
- 3) Parvo & Konio of LGN only take part (3,4,5,6)
- 4) Colour vision center is area no. V4 (fusiform gyrus)

• max. Rods ⇒ periphery (120 million)

• max. Cones ⇒ fovea (6 million)

↓
3 types :

color blindness → screening : Ishihara

Red	→	L	→	PROTANOPIA
Green	→	M	→	DEUTERANOPIA (MC)
Blue	→	S	→	TRITANOPIA

} XR → young males

COLOUR BLINDNESS

Screening thanks → ISHIHARA

Quantification test → • FARSWORTH MUNSELL 100 Hue
• ANAMALOSCOPE

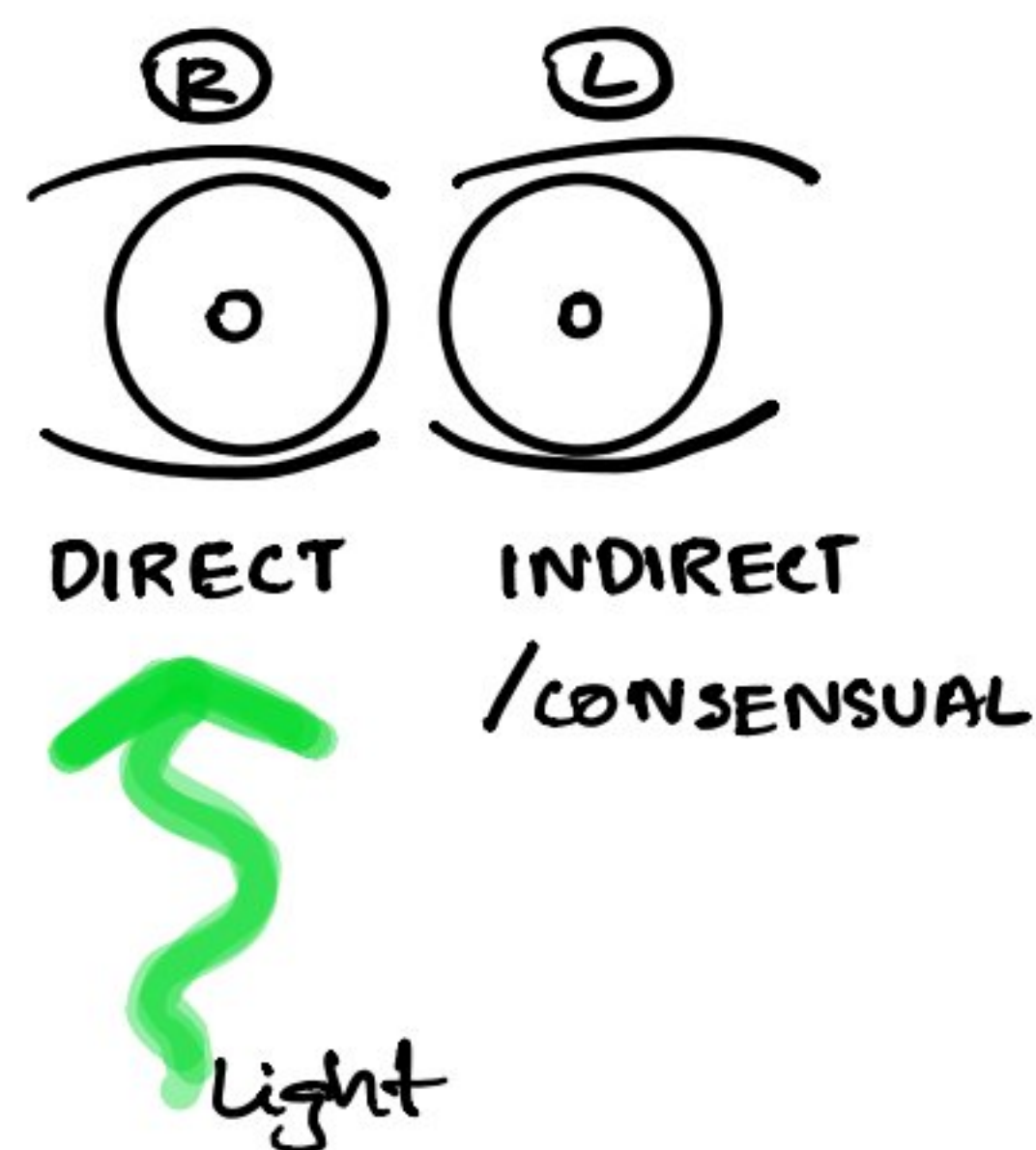
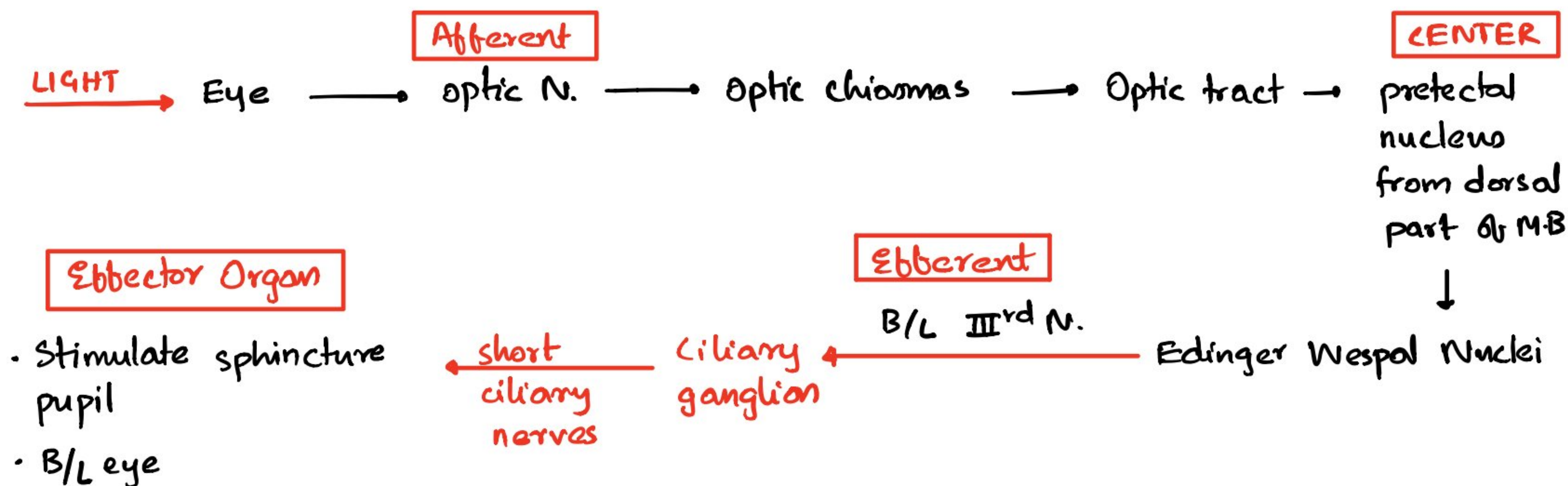
To classify → • Anamaloscope
• EDRIDGE GREEN LANTERN



Foveola → • No blue cones
• best colours appreciated = Red & Green

Principle of Ishihara test : Multiple colors are shown for color confusion

Light reflex pathway (pupillary)

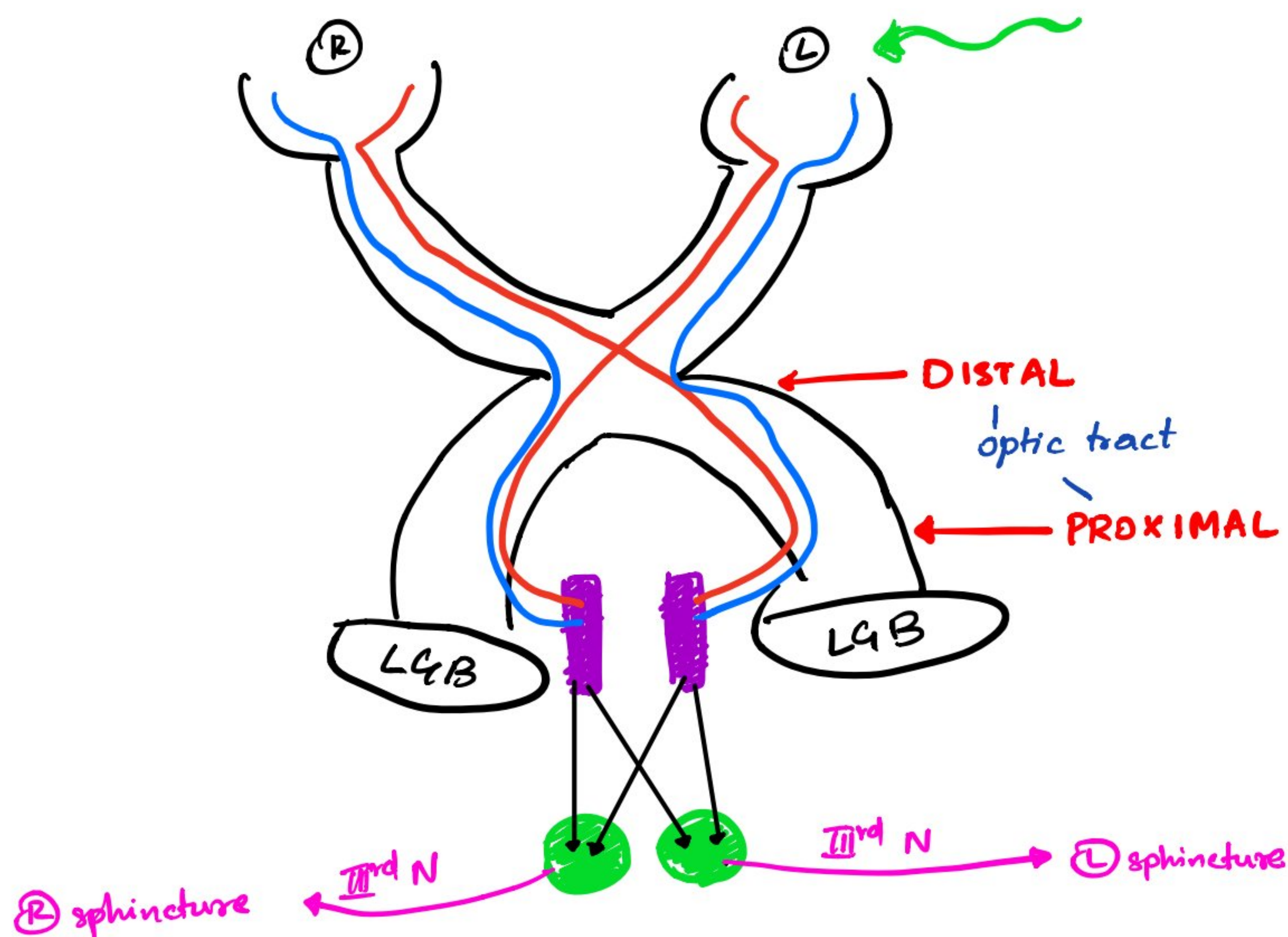


Direct Reflex
Indirect Reflex

Afferent
Ipsilateral IInd N.
Ipsilateral IInd N.

Efferent
Ipsilateral IIIrd N.
Contralateral IIIrd N.

Pupillary fibres Pathway:



PRETECTAL NUCLEUS

EDINGER WESTPHAL NUCLEUS

{ parasymp. subnucleus of 11th N @ midbrain }

lesion of LGB

Optic Radiation
visual cortex
Proximal Optic tract } Pupillary light reflex will be (N)

Edinger Westphal nucleus (@ mid brain)

↓ 11th N. { inferior division }

↓ branch going to Inferior Oblique muscle

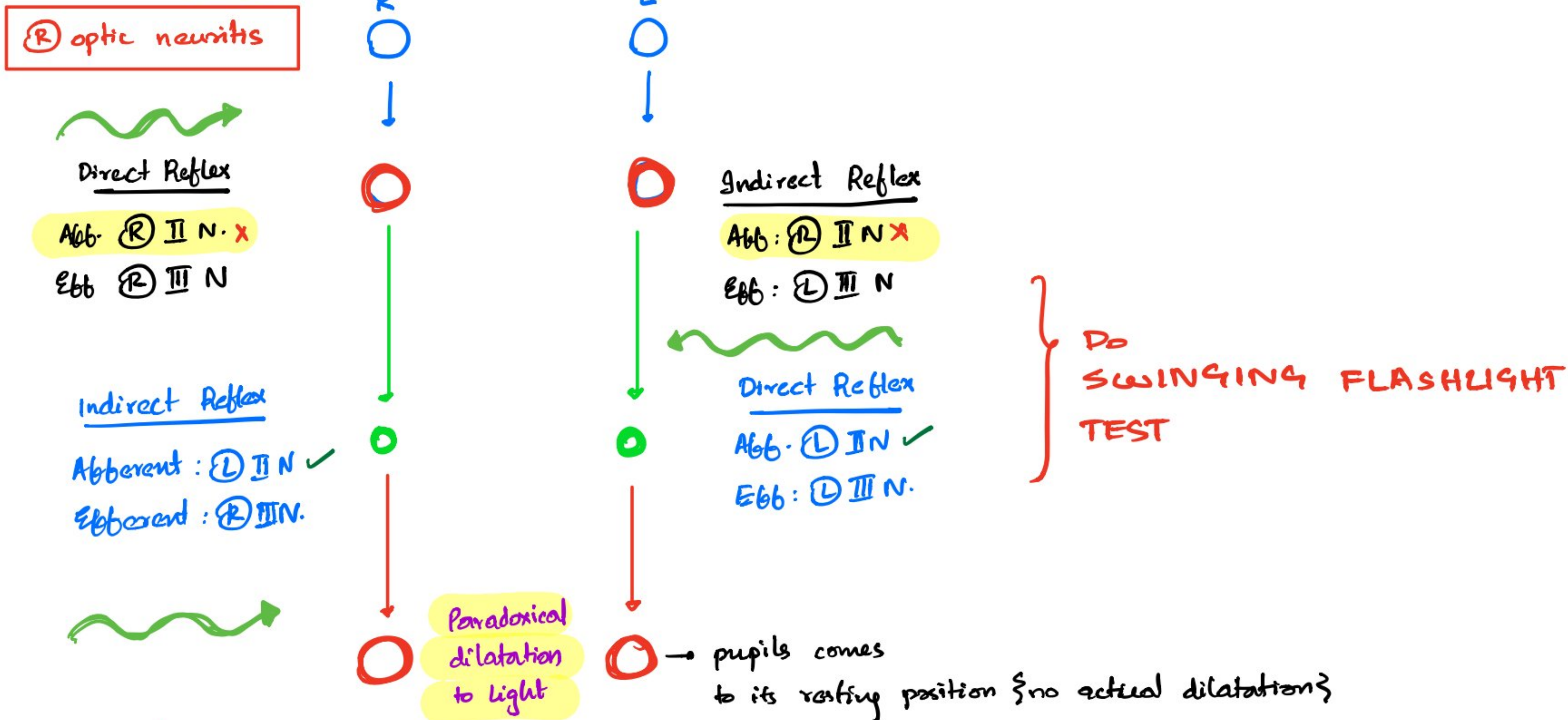
Ciliary ganglion

short ciliary N.
{ Branch V1 }

M3

RAPD {Refractive Abberent pupillary defect} / Marcus GUNN Pupil

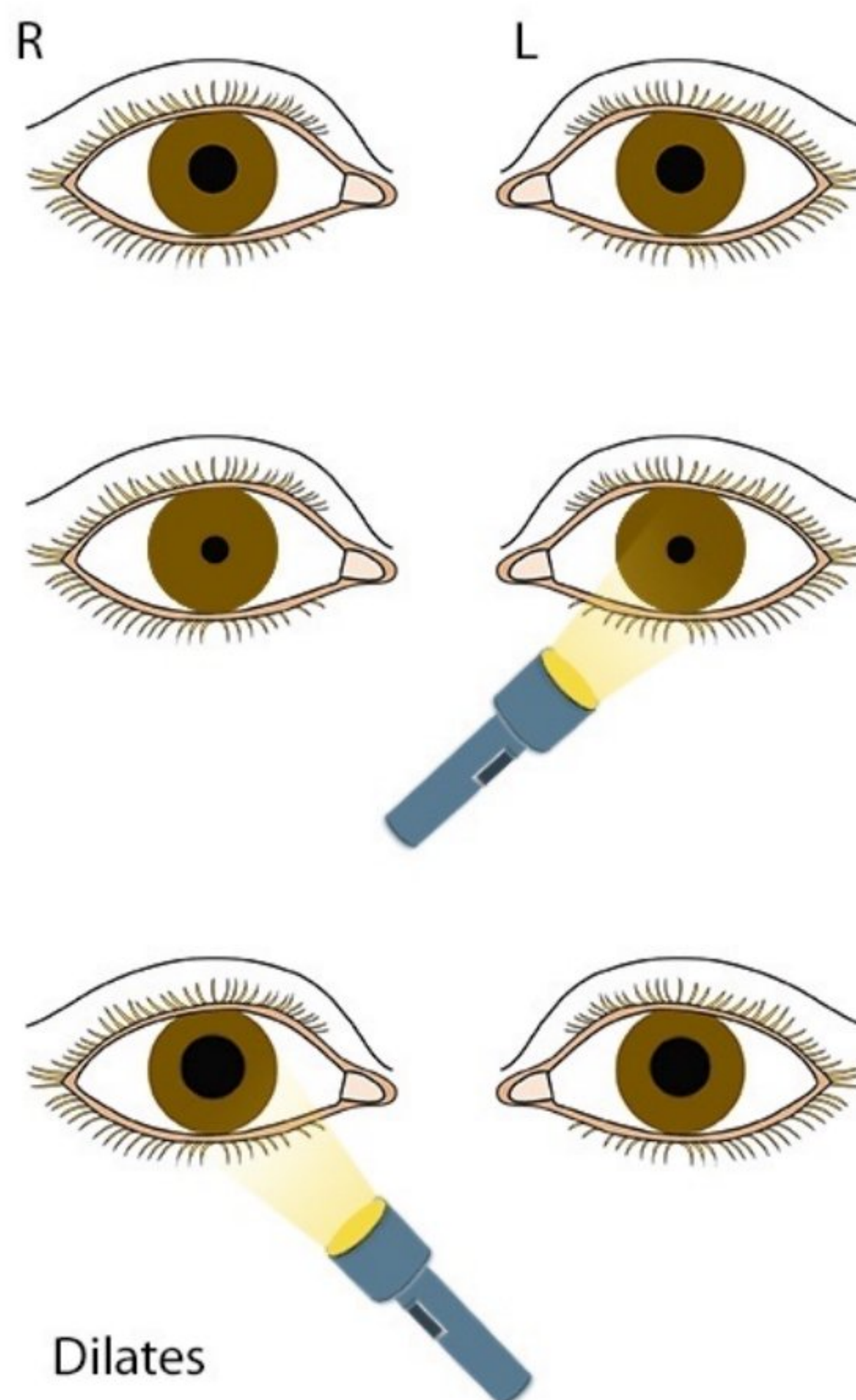
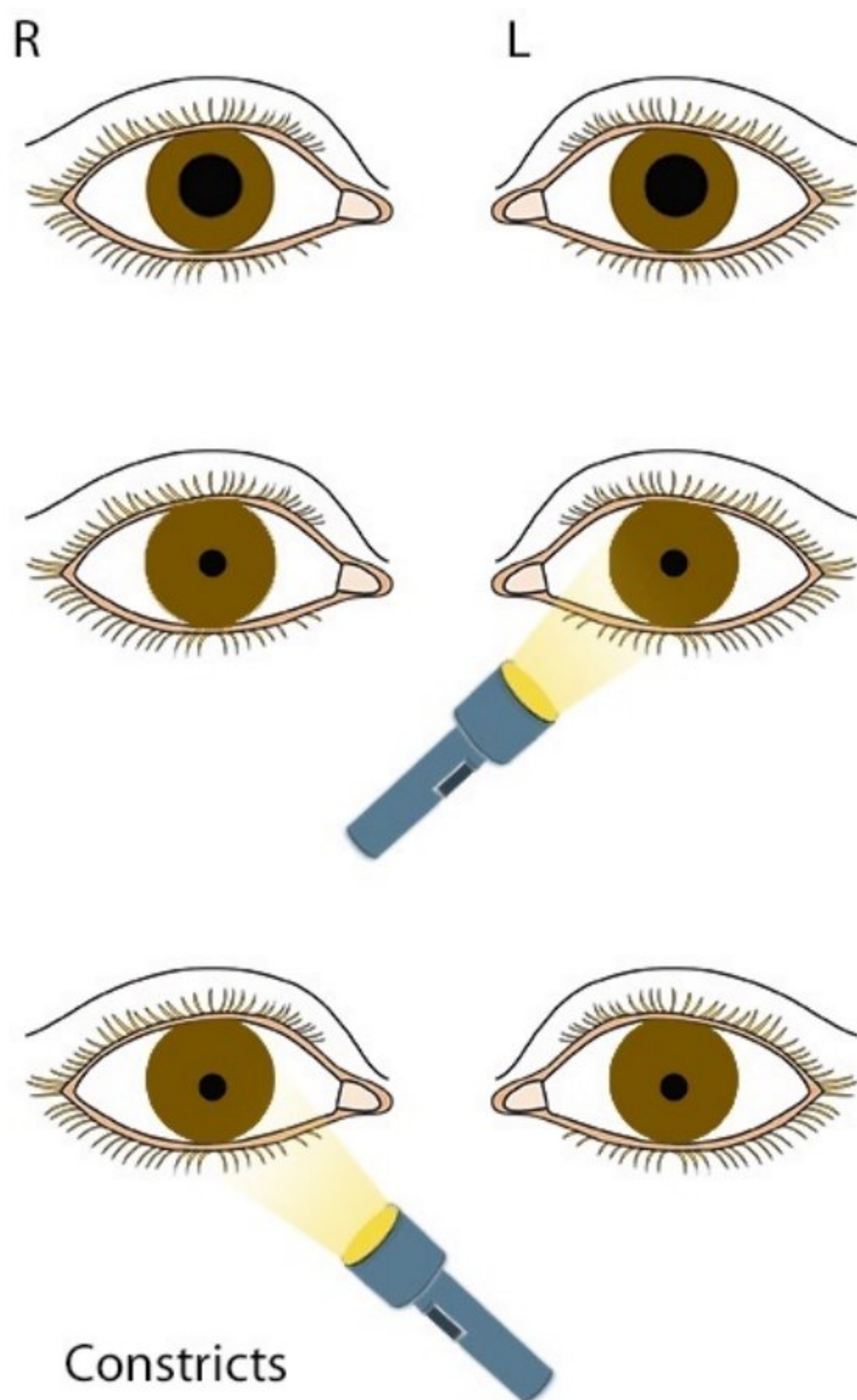
found by swinging flashlight test.



AIIMS '19

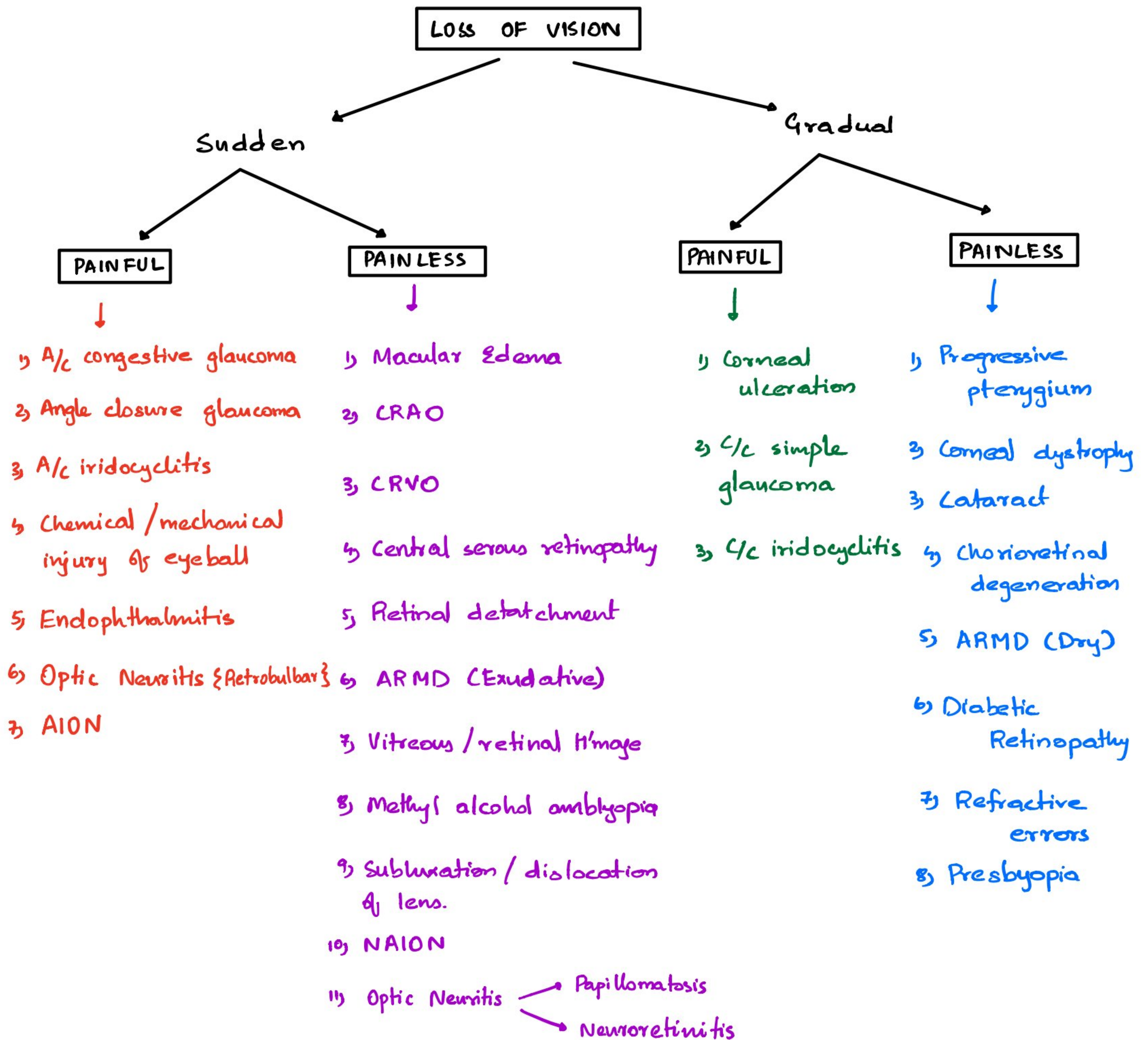
Normal

Right relative afferent pupillary defect



Q. Condition causing sudden painless loss of vision

- i) CRAO
- ii) Central serous retinopathy
- iii) Vitreous H'mage



OPTIC NERVE:

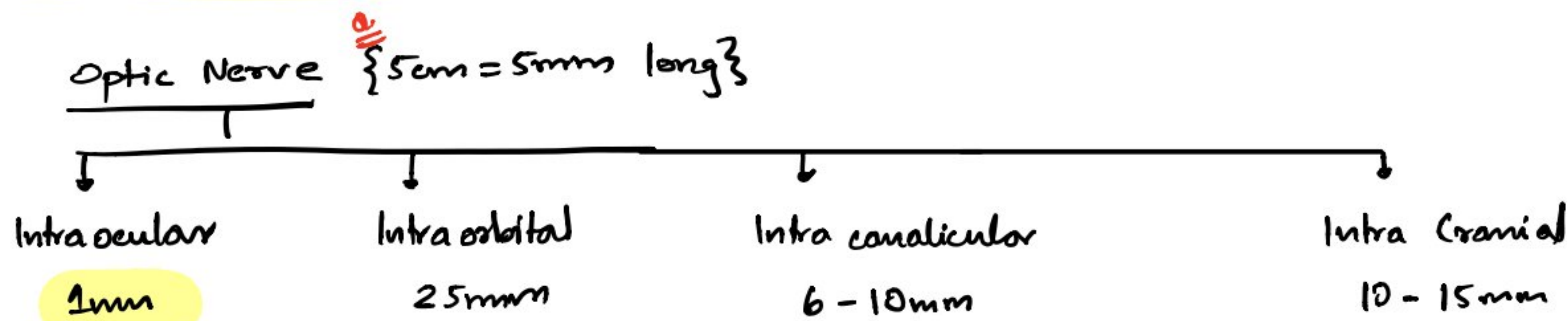
• 2nd Cranial N.

• purely sensory N.

• Axons of Retinal ganglion cells

• 1.2 million axons

• 3rd order neuron

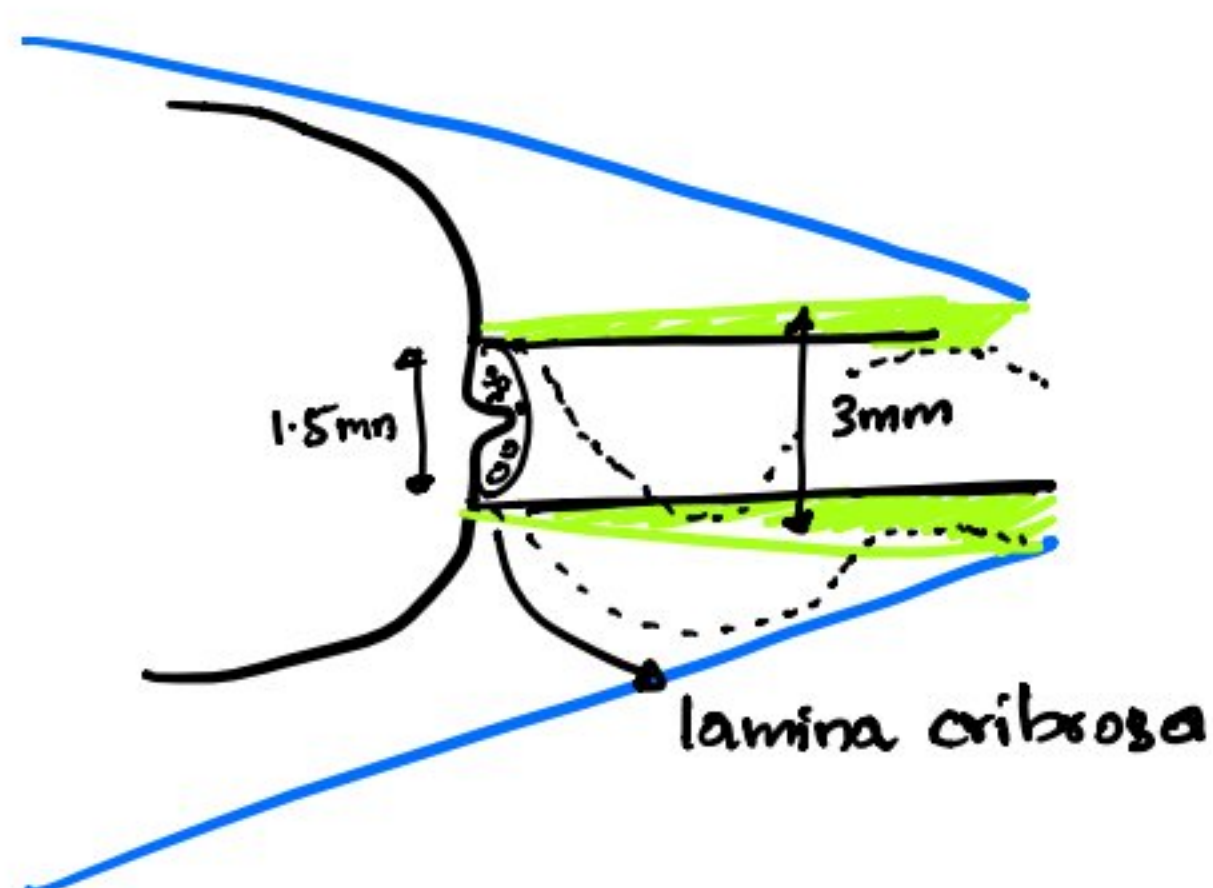


Intraocular part / optic disc / papilla / optic N. head.

• Shortest portion of O.N

• Length = 1mm

• Unmyelinated



Intraorbital Orbital

• Myelinated

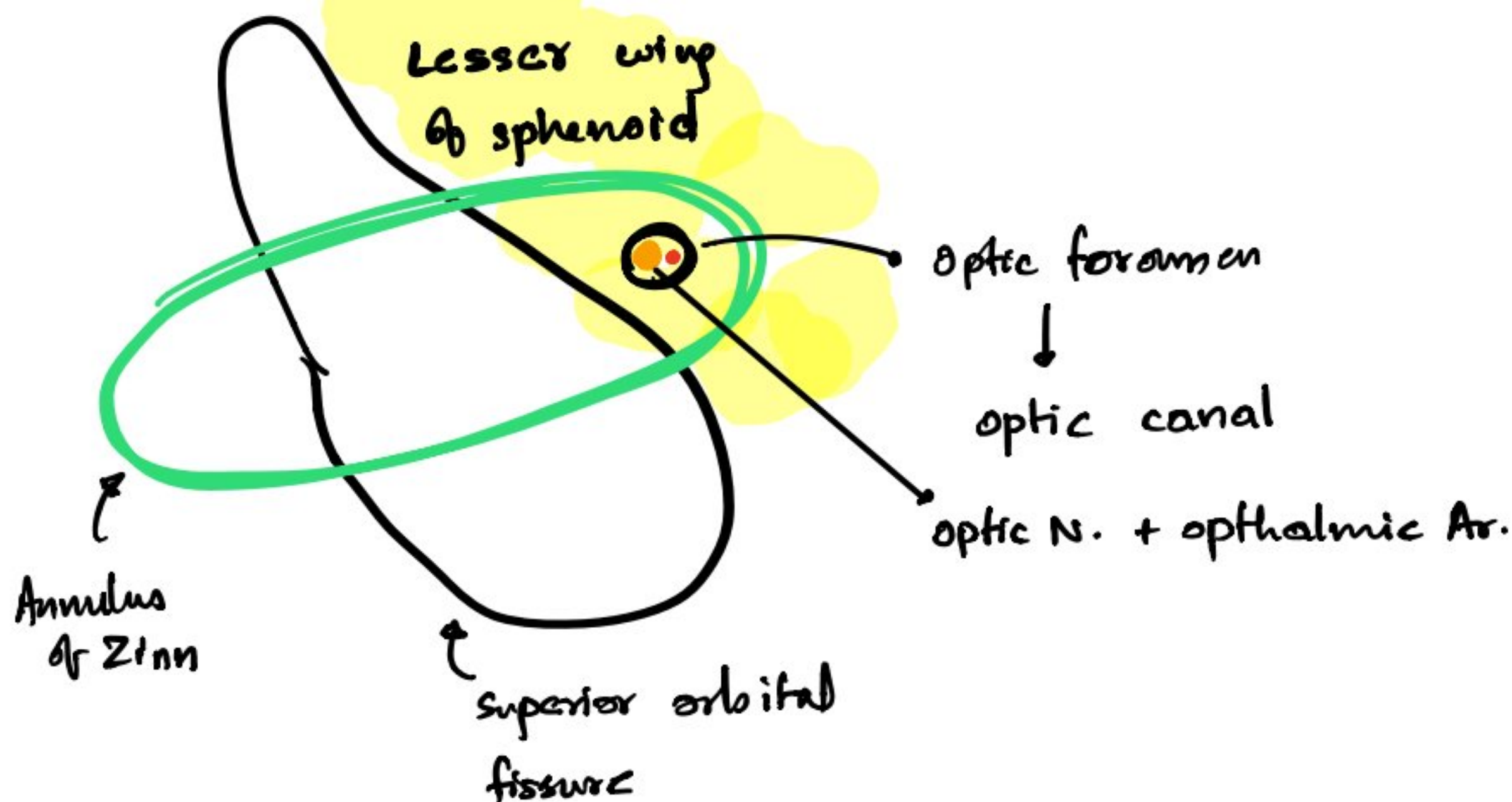
• Width = 3mm - 4mm

• Longest segment = 2.5cm

• Proptosis of $\geq 10mm$ → can only damage it.

Intra canaliculor

Trauma → canal #
→ soft tissue edema } **TRAUMATIC OPTIC NEUROPATHY**



Intra cranial Portion

Retinoblastoma → Optic N. → Brain { frontal lobe }

Frontal lobe tumor → Pressure atrophy of ON → No papilledema

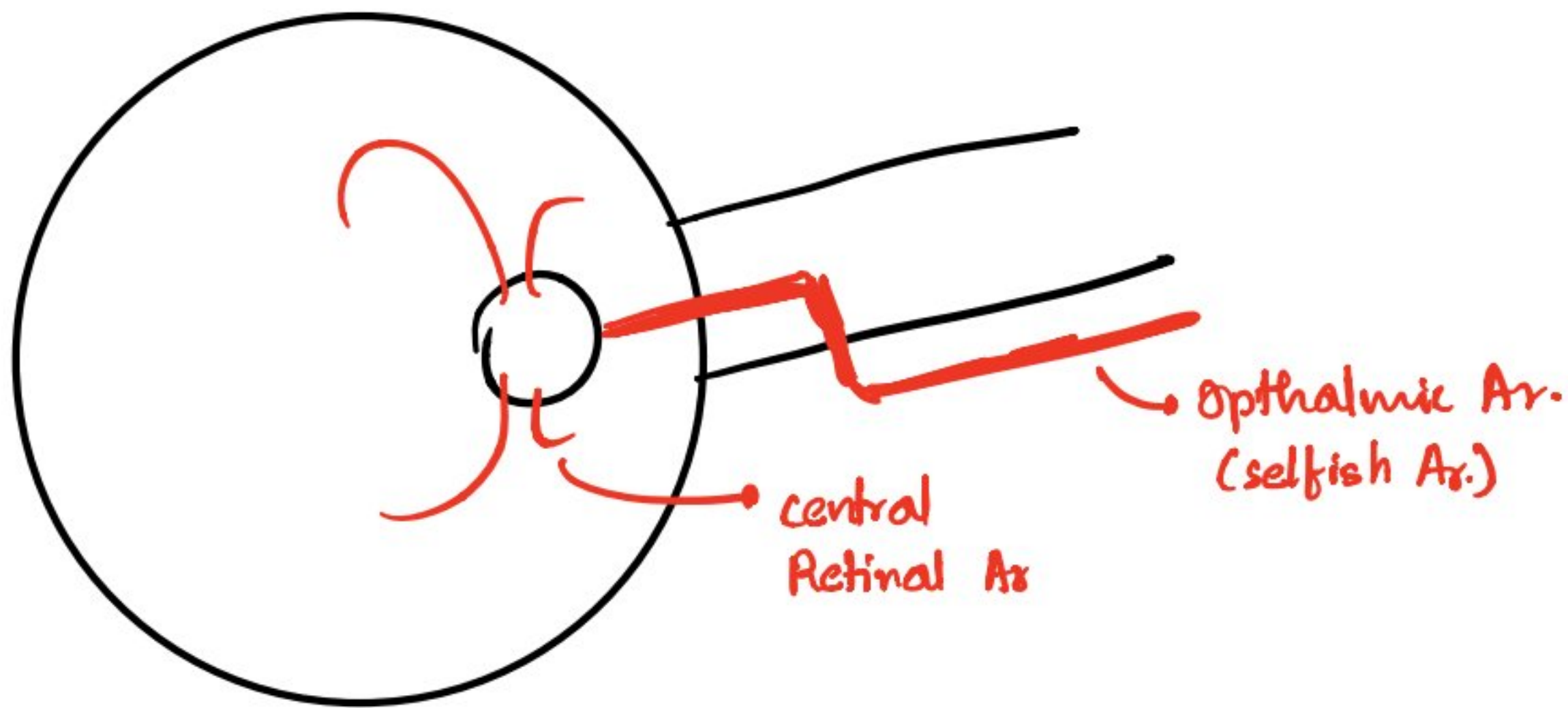
Intracranial space occupying lesion → ↑ ICP → papilledema

{ B/L optic disc edema }

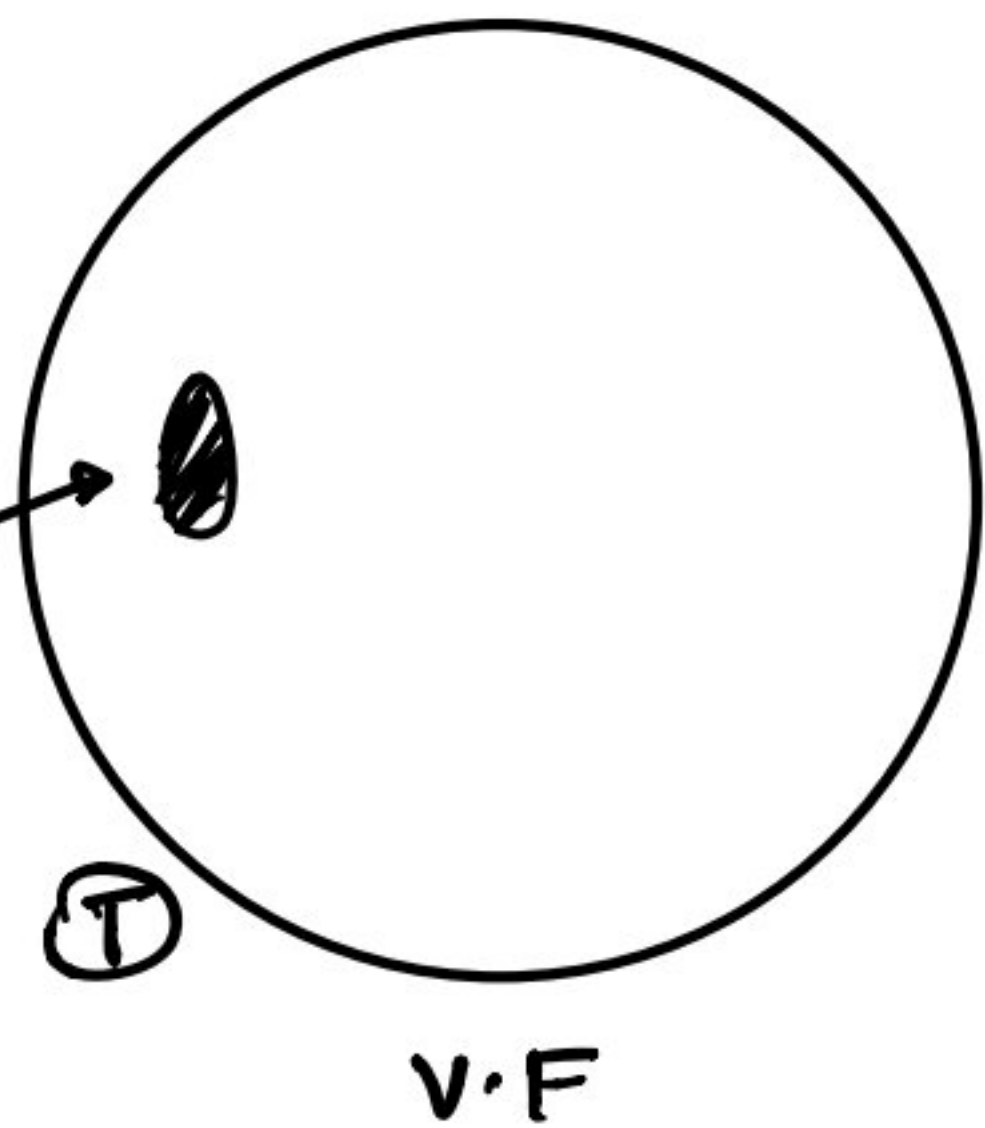
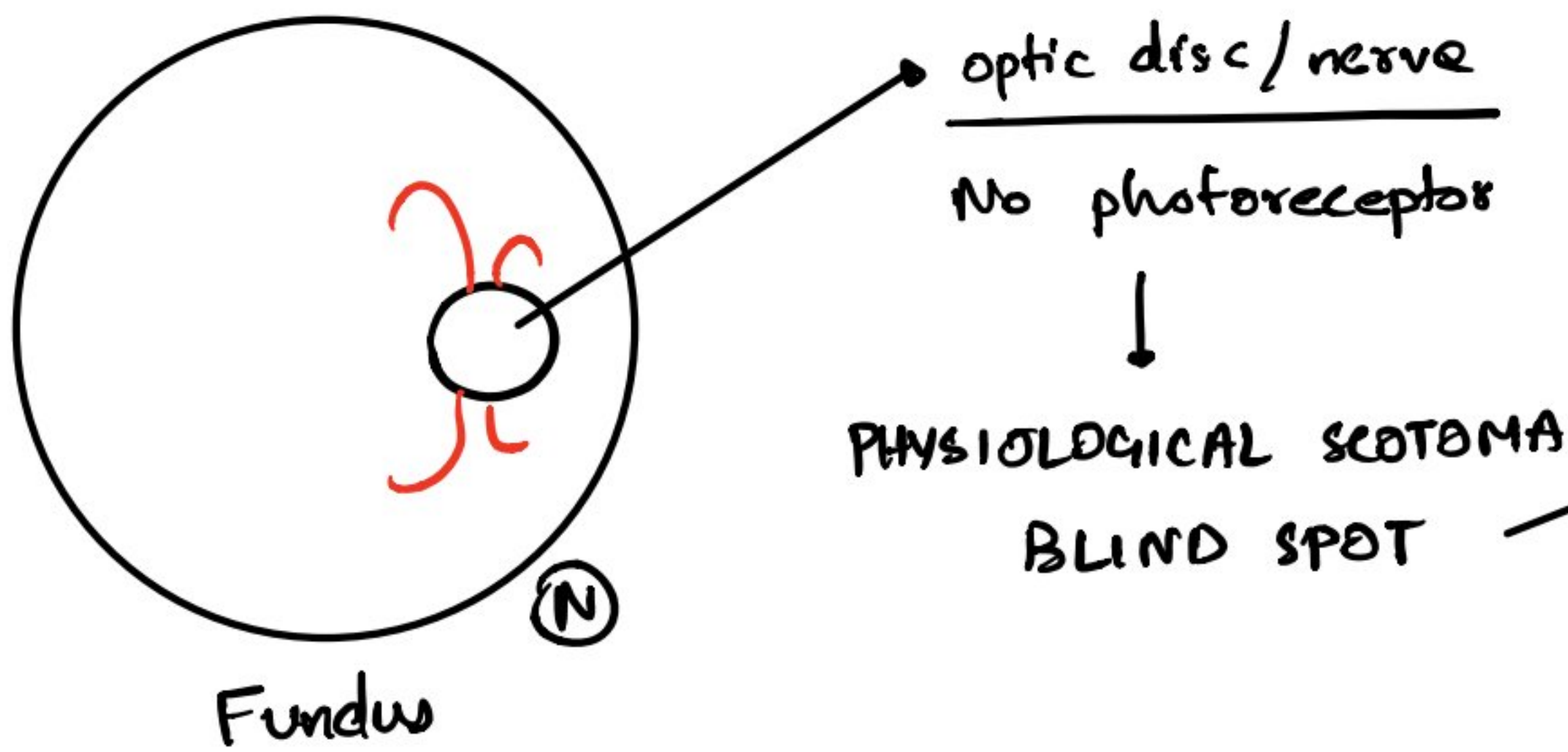
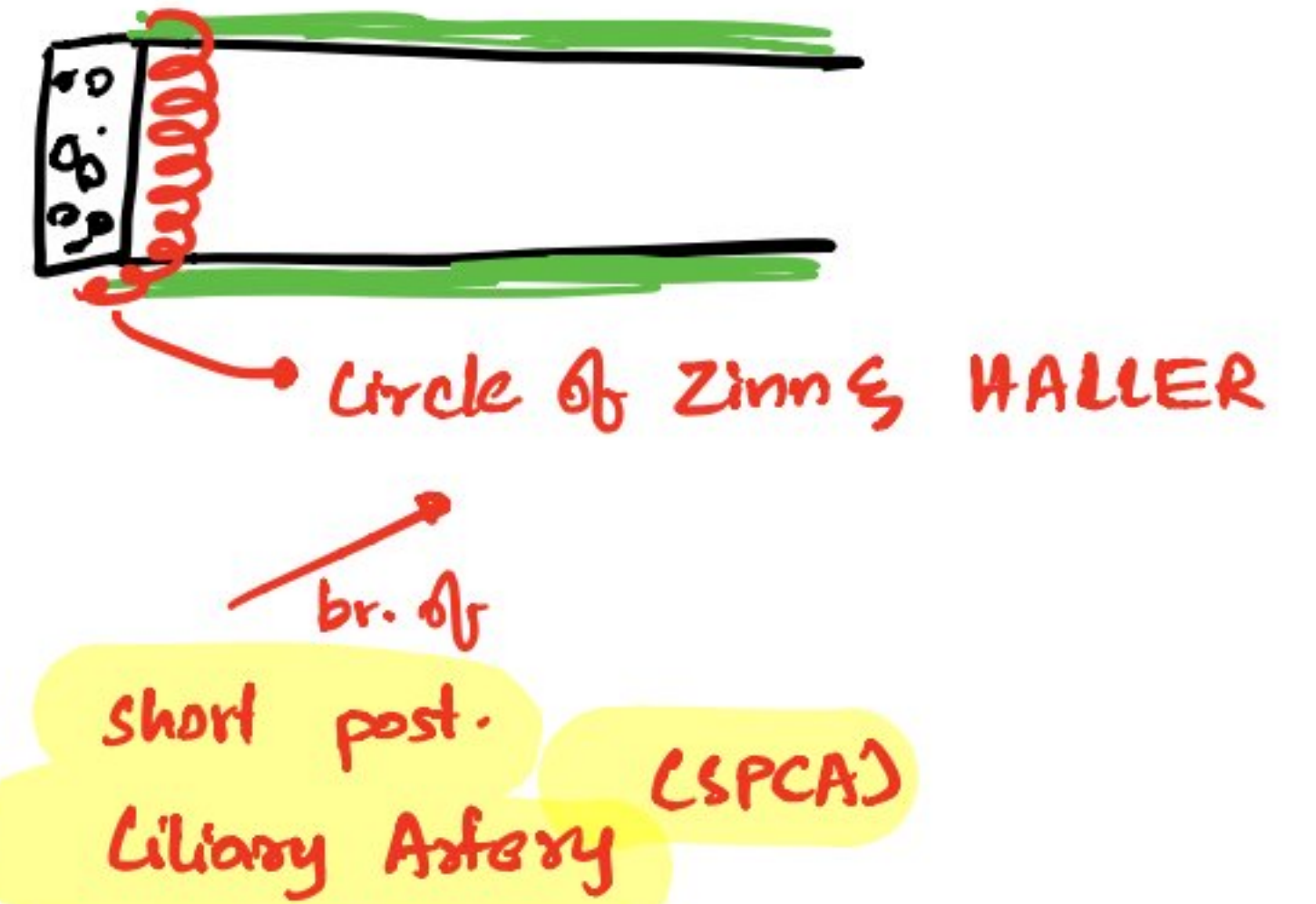
⇒ occurs only if the O.N is still intact / not atrophied

FOSTER KENNEDY SYNDROME

- Frontal lobe tumour → 1/L optic atrophy → No papilledema
- C/L disc. edema



Blood supply of Optic N.



OPTIC NEURITIS : Inflammation of Optic N.

Hallmark symptom : **SUDDEN ; PROFOUND & PROGRESSIVE** ↓se in vision

Vn
6/6 → 6/60 → 1st wk → 1/60

Clinical features :

- 1st sign → RAPD
- vision → ↓
- colour vision → ↓
- brightness → ↓
- contrast → ↓

- Red desaturation → (N)eye → Red colour dot
- Affected eye → **Desaturated dot**

- PHOSPHENES** → On movement of eye, he gets sensation of light

UTHOFF Syndrome → further decrease in vision & ↑se in body temperature

⊗ Blood supply of optic disc
↓
Central retinal Artery

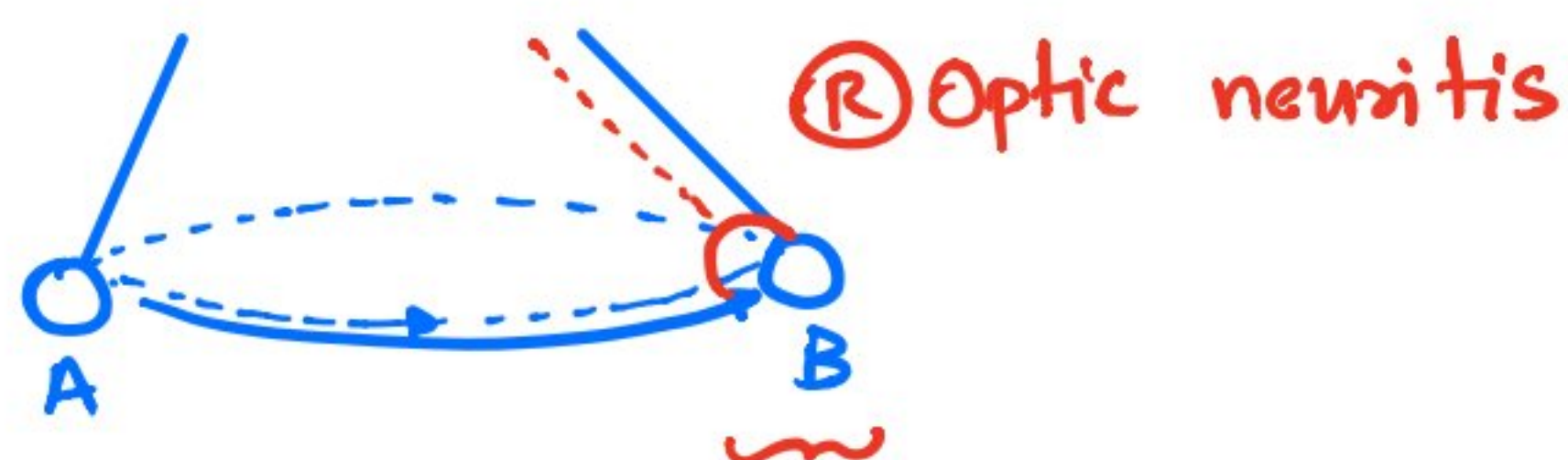
⊗ Blood supply of optic nerve
↓
short posterior ciliary artery

Q: A pt with defective vision experienced further drop in vision after hot shower / exercise

Pulfrich Phenomenon (↓ movements / depth) → STEREO ILLUSION: false sensation of depth

↓
Tested by pendulum swinging

↓ speed of conduction velocity of nerves



spatial disparity → with illusion of ellipsoid movement

• visual field defects

Types of Optic neuritis ⇒ 1st type ⇒ Retrobulbar neuritis → **PAINFUL**

• (MC) type of optic neuritis

• MCC → multiple sclerosis

• FUNDUS → (N) { Pt. sees nothing, Ophthalmologist see nothing }

on movement of eye
& d/t dural inflammation

2nd type ⇒ Papillomatosis → Optic disc inflammation

Disc edema



↓ se Axoplasmic transport



Axoplasmic stasis

(MC) type s/i children

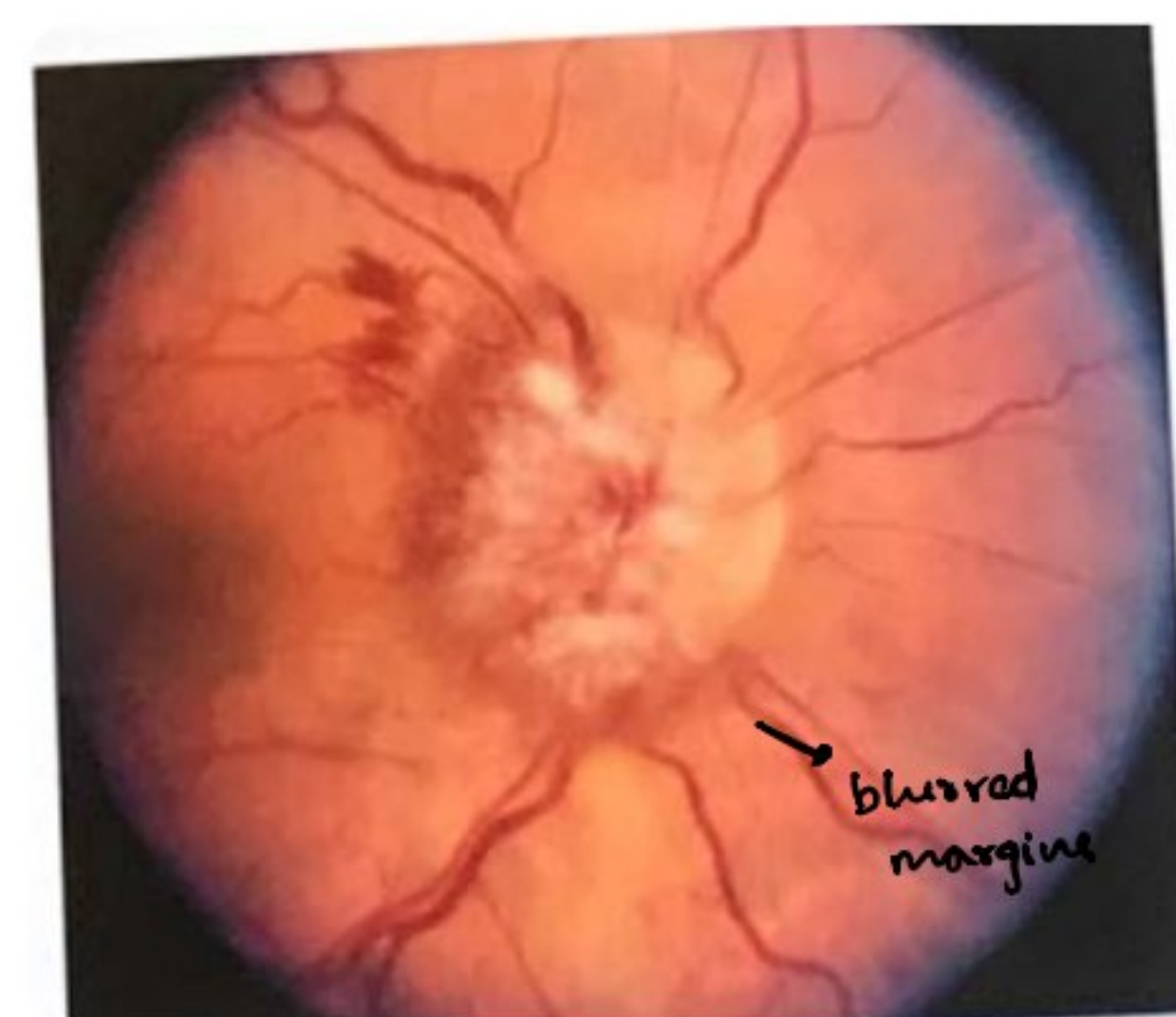
PAINLESS

3rd type ⇒ Neuroretinitis → RNFL → **PAINLESS**

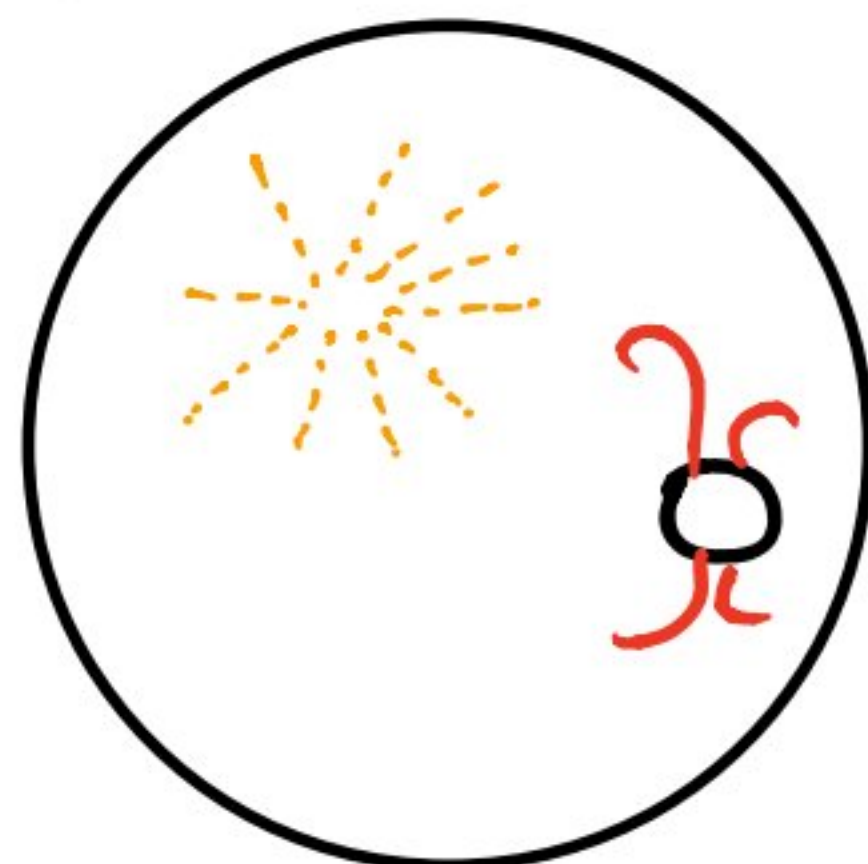
Cat scratch d/s → Bartonella Hensle

Fundus finding

• Superior Rectus
• Inferior Rectus
have some connection to duramater



Papillomatosis



MACULAR STAR

• Hard exudates in the form of radiating star.

MACULAR STAR D/D

1) Neuroretinitis

2) PIH retinopathy

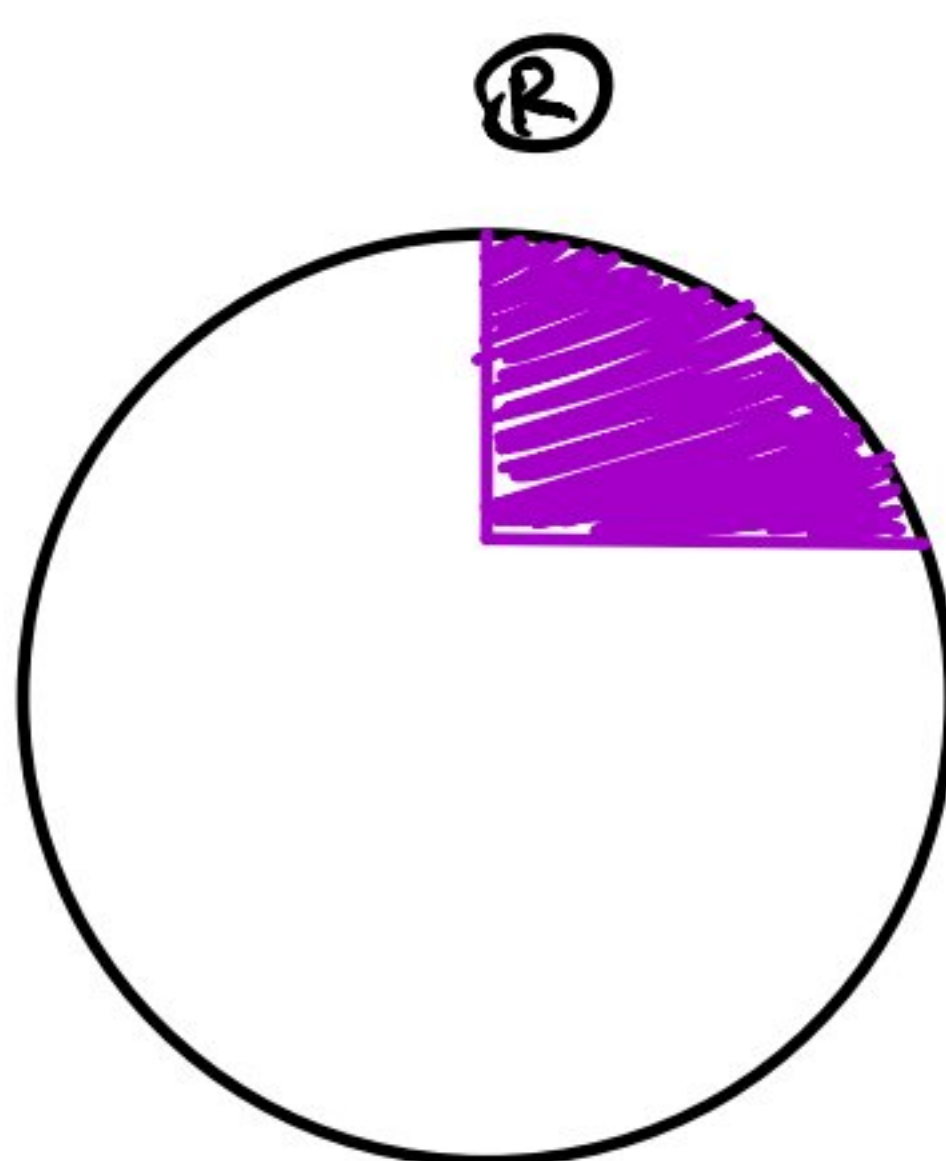
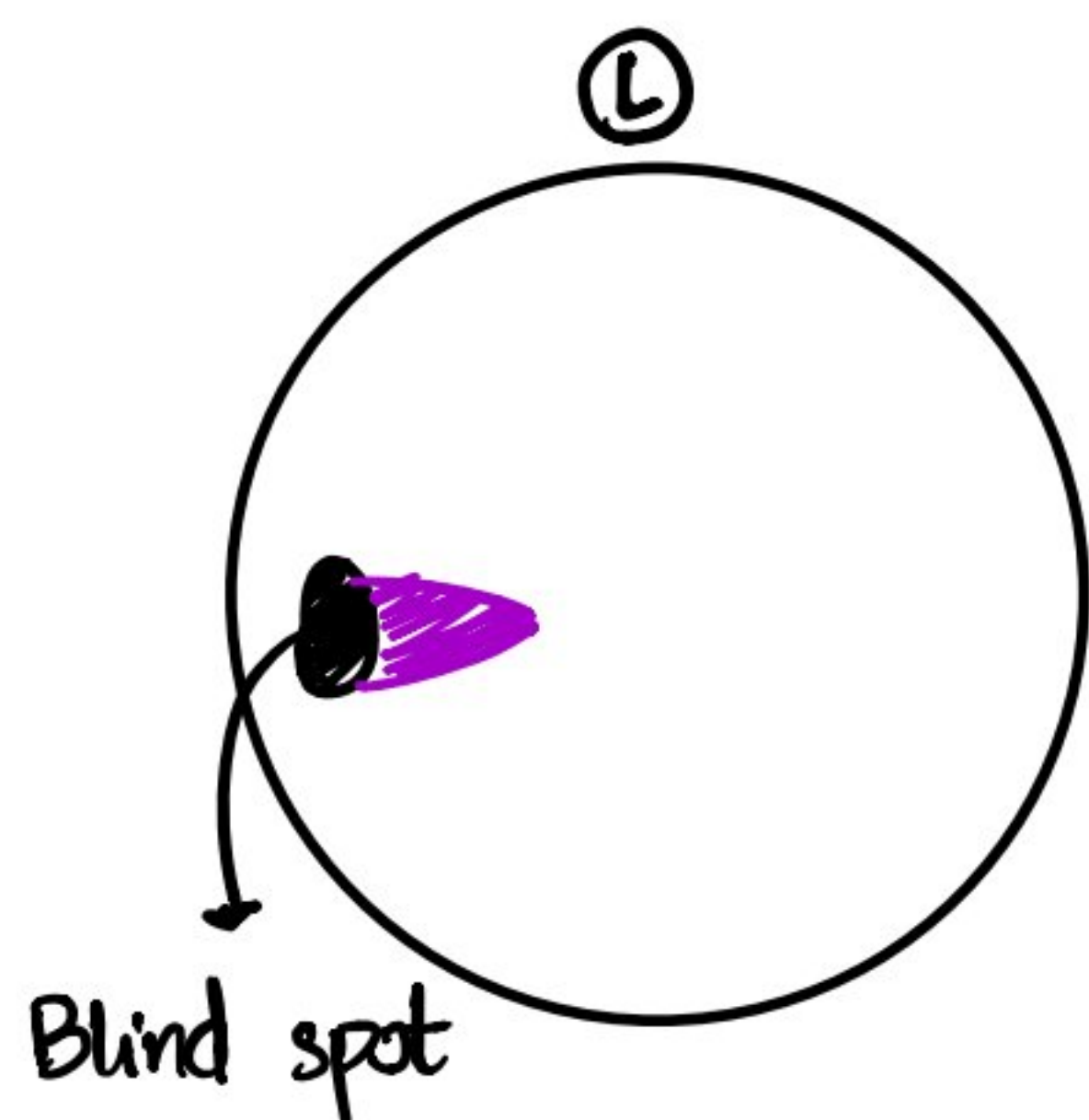
3) 4th & HTN Retinopathy

Imx OC: Contrast enhance MRI of orbit + Brain

↳ Dawson's finger: MS

Visual field defect → CENTROCAECAL SCOTOMA or Centra Scotoma

Optic Neuritis ⇒ doesn't affect all the fibres of optic N. gives Centrocaecal scotoma



Ⓐ centrocaecal scotoma + Ⓑ supertemporal quadrantanopia } Junctional

Causes of Optic Neuritis

↳ Demyelination → MS
↳ MCC
↳ Young females
↳ Painful eye movement
↳ Uhthoff phenomenon

Rx: I.V steroids × 3 days
Oral steroids × 11 days

Other cause

↳ Devic d/s

↳ Leukodystrophies s/i a) Krabbe's d/s

b) Schiller's Adrenoleukodystrophy

↳ Infection → Bacteria → Lyme's d/s, Syphilis, Cat Scratch, Cryptococcus.
↳ Viral → Meningitis in AIDS/TB

↳ Whooping cough, Measles, Mumps, VZV

↳ Heritable → Lebers Hereditary optic neuropathy {Mitochondrial inheritance}
pupil Rx. is paradoxically Ⓐ

• Typically s/i M > F, 3rd decade

Initially → U/L visual loss

↳ fellow eye involved in 2 months → B/L optic atrophy

↳ Ischemic
SPCA → AION
N-AION

Arteric Ischemic Optic Neuropathy

Old age pt. present with giant cell arteritis

↳ headache

• Jaw claudication

• scalp tenderness

• Inflamed temporal artery

} → sudden painful loss of vision.

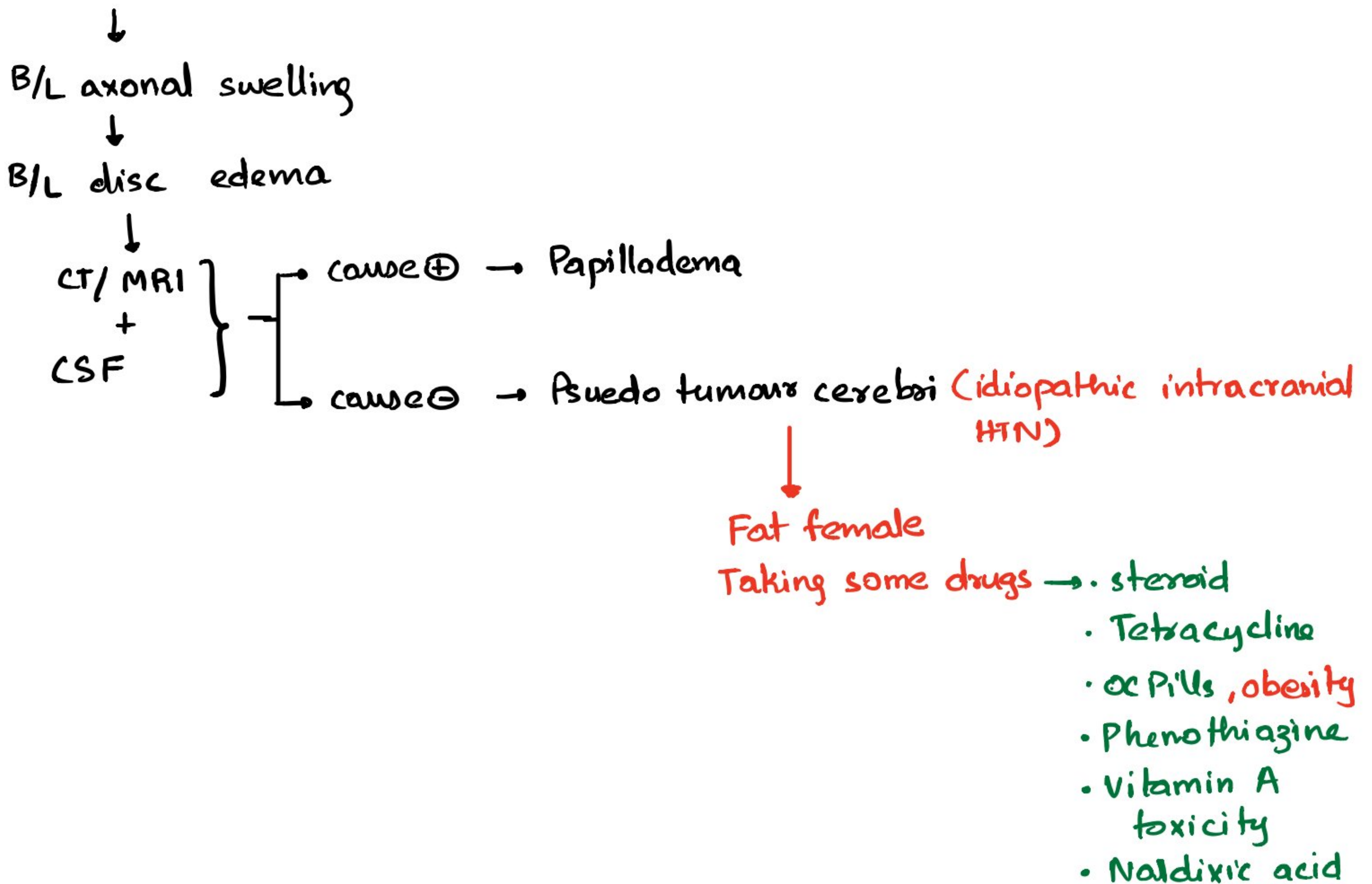
Rx → I.V steroids (Hydrocortisone) for 3-4 day along w oral prednisolone

Non arteritic Ischemic optic neuropathy

- Hypertension
- nocturnal hyper perfusion
- sleep apnoea → Painless loss of vision in morning
- Altitudinal defect → either superior / inferior field defect

Rx → Stop smoking, check PCV & fibrinogen | Manage BP & underlying cause

5) Intracranial cause (↑CSF)



investigation :: ↑CSF Pressure
• Disc edema ⊕
• CT/MRI ⇒ ⊖

Rx :: Stop the drugs
• Wght reduction, Furosemide
• Optic Nerve Fenestration surgery
• Ca Anhydrase ⊖
f/b Serial LP

Rx : Systemic Steroids $\Rightarrow$ 14 day (3+11) PROTOCOL

- 1st three day $\rightarrow$ IV MP $\S$ Methyl prednisolone
- 11 days : Oral Prednisolone 1-2mg/kg/day $\rightarrow$ only oral steroid can lead to recurrence
• So it's C/I

Anterior Ischemic Optic Neuropathy

cause: Occlusion of short posterior ciliary artery (SPCA)

Presentation :

Types $\begin{cases} \rightarrow \text{NAION} \\ \rightarrow \text{AION} \end{cases}$

NAION $\rightarrow$ MC type

- MCC $\Rightarrow$ Atherosclerosis / HTN
- Presentation $\rightarrow$ sudden painless loss of vision
- Optic neuritis $\rightarrow$ RAPD

Fundal Finding



NAION

- Disc edema
axonal swelling
 $\downarrow$
axoplasmic stasis

Hyperemia of disc

Mx : Rx underlying cause

2) A-AION

- MCC $\rightarrow$ Giant cell arteritis
- sudden Painful loss of vision
- Amaurosis fugax $\S$ Transient visual blackouts $\rightarrow$ NIMHANS occur d/t occlusion of Retinal artery
- Jaw claudication $\S$ Pathognomic
- Scalp tenderness
- a/w Poly fibromyalgia Rheumatica

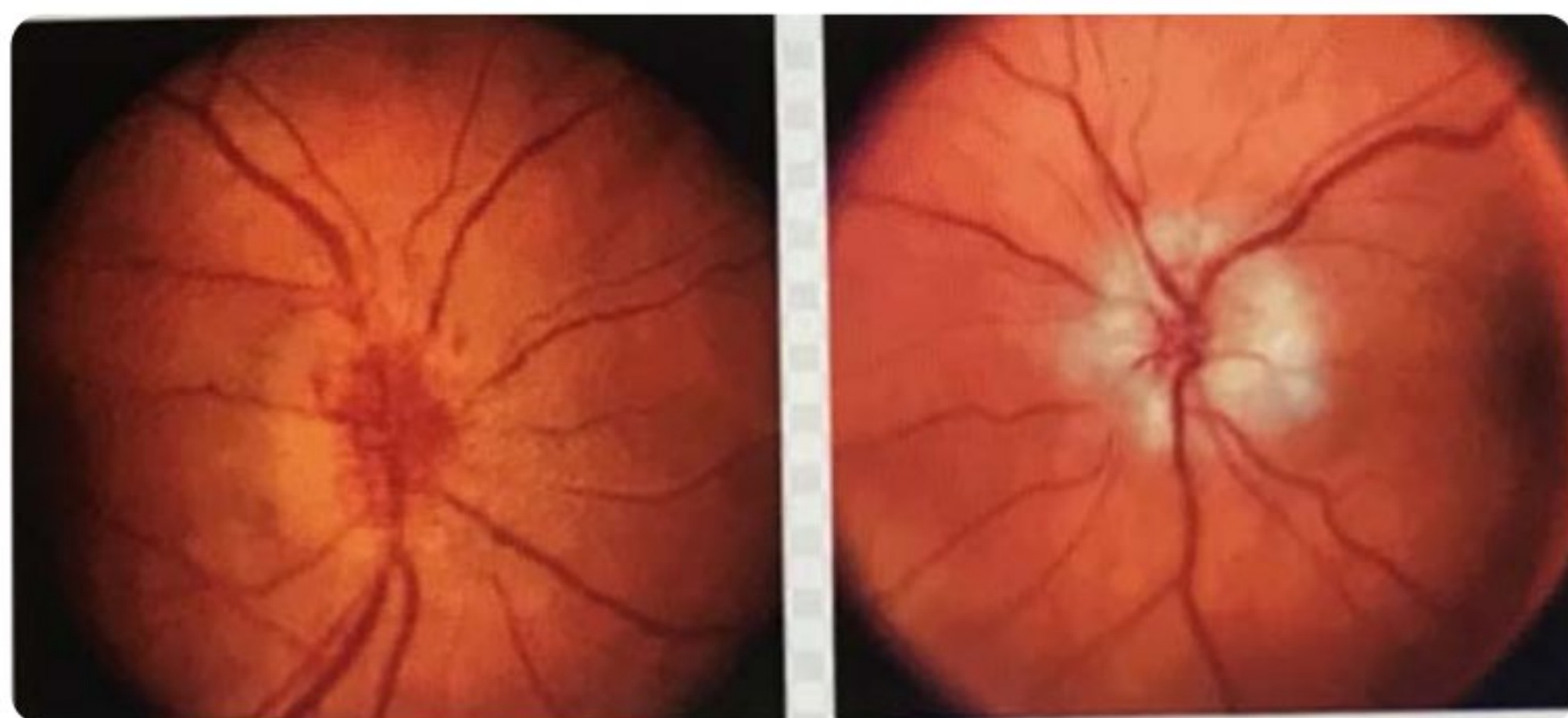
Fundal Findings



A-AION

Axonal swelling
 $\downarrow$
Axoplasmic stasis
 $\downarrow$
Disc edema

PALE DISC EDEMA



NAION/ A-AION

Management of A-AION ::

Invest : CRP , ESR ↑ sed

Gold std : Temporal Arterial Biopsy
of 3mm length skip lesion?

Rx : Systemic steroids

1st 3day of IV MP

↓ f/b

11days oral Prednisolone

To protect involved & fellow eye

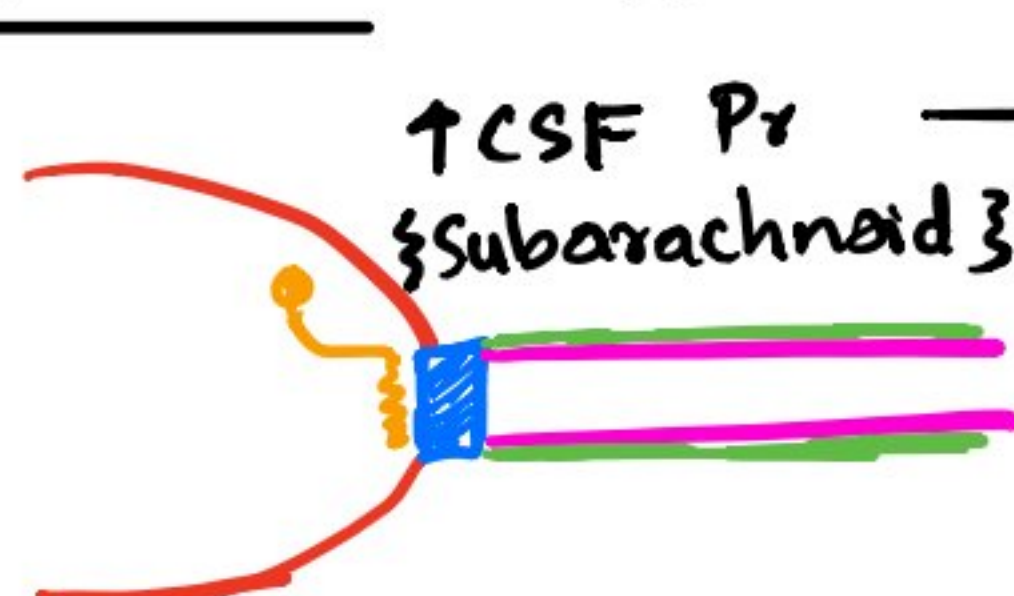
Visual field defect in AION



Inferior Altitudinal Hemianopia

superior arching fibres are first affected

PAPILLEDEMA : B/L disc edema d/t ↑ ICP (CSF Pr.)



↑ axoplasmic stasis

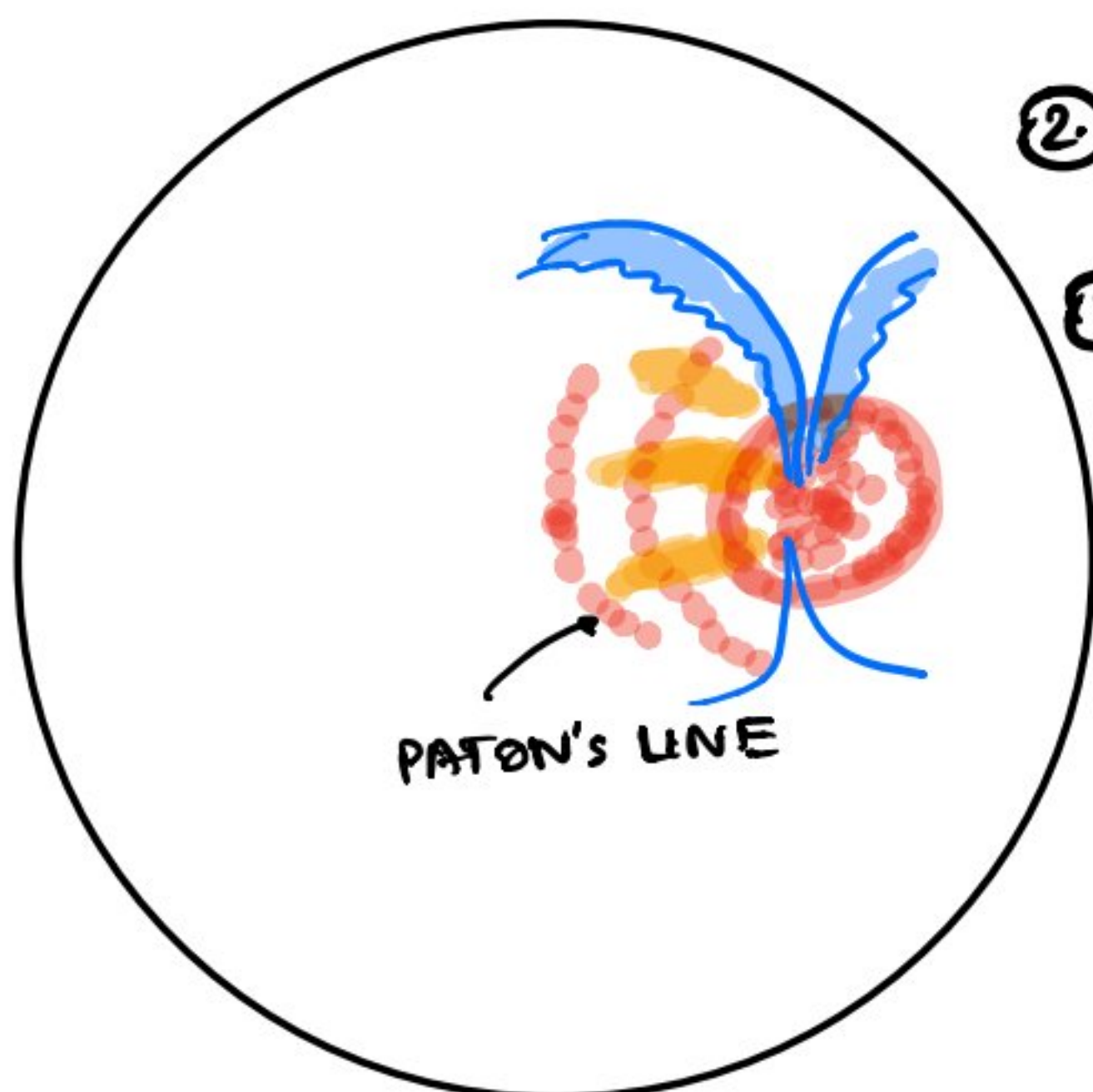
Disc edema

1st sign ⇒ ① Blurring of nasal disc margin

② Obliteration of Cup

③ Disc surface elevation

④ Peripapillary Edema → PATON'S LINE



1) Dilated & Tortuous vein {d/t compression}

2) Disc H'mage & d/t back pressure?

3) Ischemia → Cotton wool spots

4) Spontaneous Venous pulsation → Absent (Normal condition)

9f ④: Never papilledema Dx of exclusion

unmanaged

NEETPG

C/c Papilledema → "Champagne cork" app

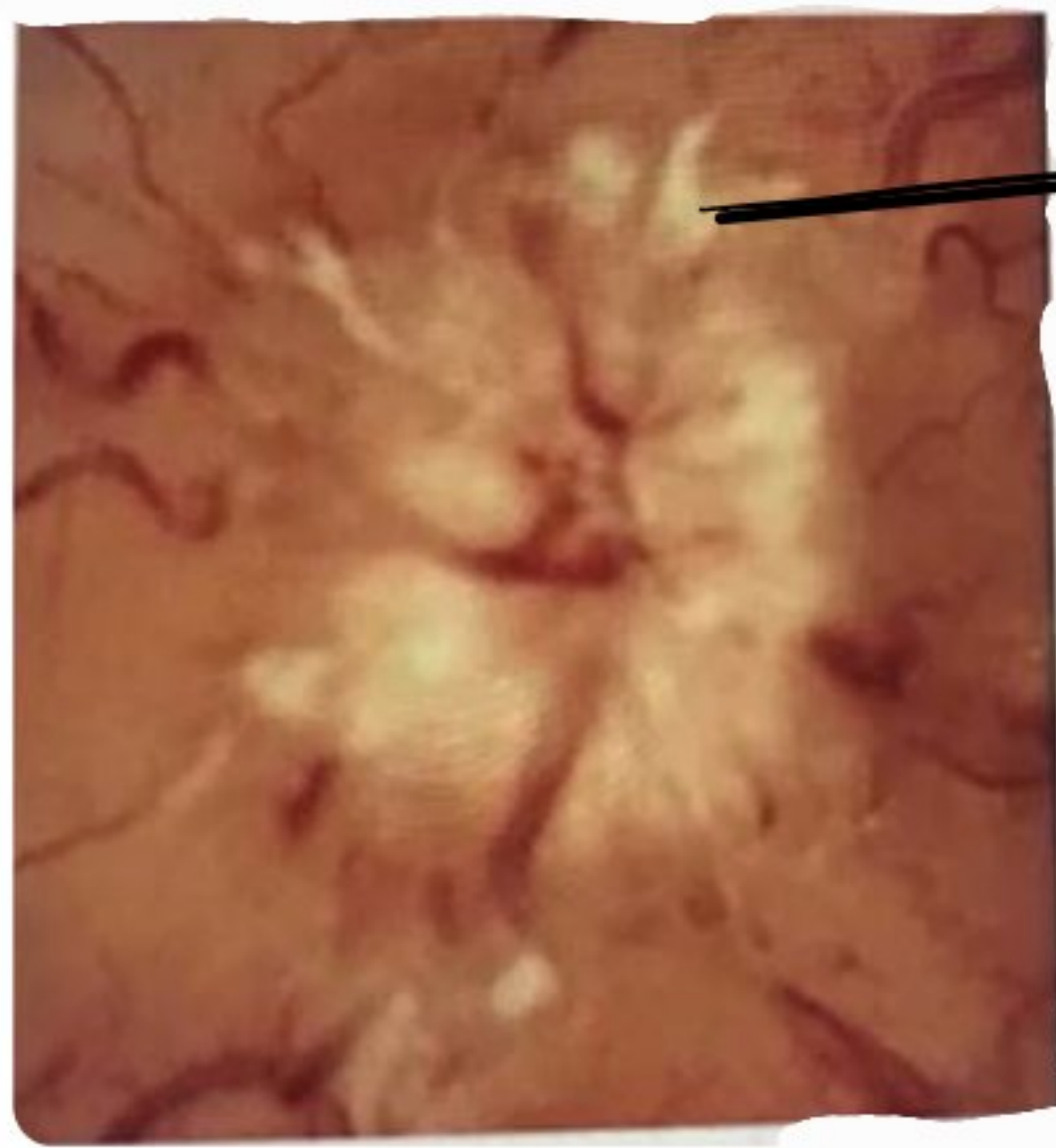
↓ on Mx
2° Optic Atrophy

Fundus finding in Papilledema

PAPILLEDEMA



A/C papilledema



C/c papilledema - Shampaigne cork sign

Symptoms:

- 1) Headache (worse on awakening)
- 2) Projectile vomiting
- 3) Diplopia (VIth N Palsy → LR palsy → Paralytic squint)
- 4) Vision acuity (N)
Blurring (+)
- 5) Amaurosis Fugax → fleeting episodes of loss of vision & d/t retinal vein compression

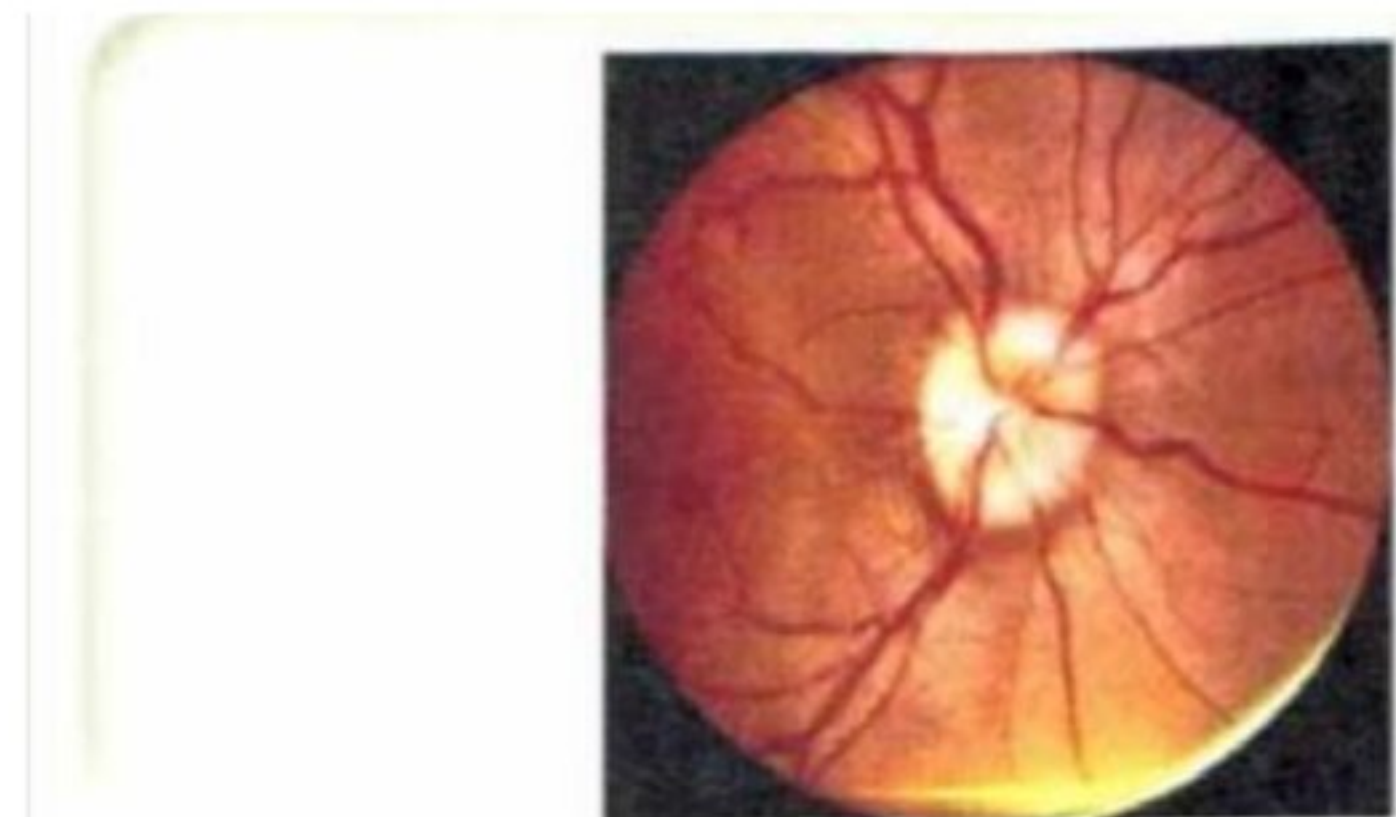
Visual field defect : B/L enlarged blind spot

↳ C/c stage: concentric constriction of visual field

Pupil → light reflex (N)

Mx → ↓ ICT

Q. Diagnose the Disorder shown in the Photograph



- a) Optic Neuritis
- b) Optic atrophy
- c) Papilledema
- d) Tumours

Ocular Ischemic Syndrome

- d/t C/c ocular hypoperfusion 2° to carotid artery stenosis (90% stenosis)

Risk factor:

- i) 0
- ii) Old age [60-90yr]
- iii) Smoking
- iv) HTN
- v) DM
- vi) Cholesterol

C/F:

- 1) Amourosis Fugax (Transient Retinal ischemic attack)
- 2) Retinal artery occlusion
- 3) TIA/stroke
- 4) Asymmetric DM Retinopathy
- 5) No papilledema

Pseudo tumour Cerebri § idiopathic / benign intracranial HTN?

- Papilledema ⊕
- ↑sed CSF pressure
- No cause
- CT/MRI Brain = (N)

Normally: CSF pressure ↑sed → Ventricle size ↑sed
 But IHH: " " " → Ventricle size (N) / ↓sed

Hallmark of this condition → Raised CSF opening pressure on LP
 LP CSF analysis = (N)

Risk factors: Perimenopausal fatty ♀

- Taking some drugs →
- Steroid
 - Tetracycline
 - OC Pills, **obesity**
 - Phenothiazine
 - Vitamin A toxicity
 - Nalidixic acid
 - Nitrofurantoin

"old fatty lady"

S
T
O
P

V
A
N

Visual field defect → B/L enlarged blind spot

Rx: Stop the drugs,
 weight reduction

Medical Mx : F → Furosemide
 A → Acetazolamide → Carbonic Anhydrase
 T → Topiramate {antiepileptic}
 s/e wght loss is used.

Surgical Mx → i) VP Shunts
 ii) ON sheath fenestration → CSF release

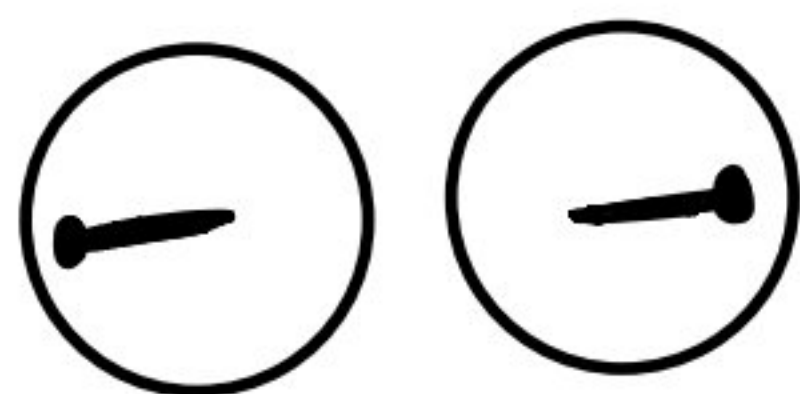
TOXIC OPTIC Neuropathy [Toxic Amblyopia] → loss of vision d/t toxic drugs

Drugs causing "CLAIMED"

- ① Chloramphenicol / cyanide of Tobacco → Tobacco Amblyopia
- ② Lead
- ③ Alcohol = Ethanol (2nd MCC)
- ④ Isoniazid
- ⑤ Methanol (Formic & formaldehyde) → ^{JIPMER} affects the ganglion fibre layer
- ⑥ Ethambutol
- ⑦ Deficiency of Vit B1 & 12

Centrocecal Scotoma → Papillomacular bundles are damaged

i) Defb. of Vit. B12 → B/L



ii) Tobacco Amblyopia

OPTIC ATROPHY

	PRIMARY	SECONDARY	CONSECUTIVE
Disc	No antecedent optic disc swelling chalky white well defined margins	after optic disc swelling Dirty gray-white ill defined margins	Due to D/S of inner retina & blood supply waxy pallor of Disc less sharp margins
Optic Cup	Shallow Saucer Shaped	Filled up	Normal
vessels	Normal	peripapillary sheath	Attenuation
Surrounding Retina	Normal	hyaline bodies / Rosen	pathology (+)

causes

Optic neuritis
Tumor
Trauma
Hereditary
Toxic



Papilloedema
Papillitis
AION



CRAO/CRVO
Healed vasculitis
Pamietal photocoagulation
Retinitis pigmentosa



KESTENBAUM INDEX - Ratio of small to large blood vessels on disc surface
Optic (N) Atrophy $\frac{10}{<7}$

HEREDITARY OPTIC NEUROPATHY

KJER

AD inheritance
B/L insidious visual loss
+ SNHL

BEHR

AR
Ataxia
Nystagmus
MR

WOLFRAM (DIDMOAD)

AR
Diabetes insipidus
Diabetes mellitus
Deafness

LEBER'S HEREDITARY OPTIC NEUROPATHY (LHON)

- Mitochondrial
- maternally transmitted
- M/c manifestation in males
- U/L
- Sudden painless loss of vision
- Peripapillary Telangiectatic microangiopathy
- pseudo papillitis
- primary optic atrophy
- central or centrocecal scotoma
- (N) pupillary reactions

Causes of Pseudo-papilloedema

- LHON
- Methanol poisoning
- Near vision loss
- Optic disc atrophy
- peripapillary myelinated (N) fibres

C/F: 2nd - 3rd decade → Young adult ♂

• Hyperemic optic disc & telangiectatic capillaries

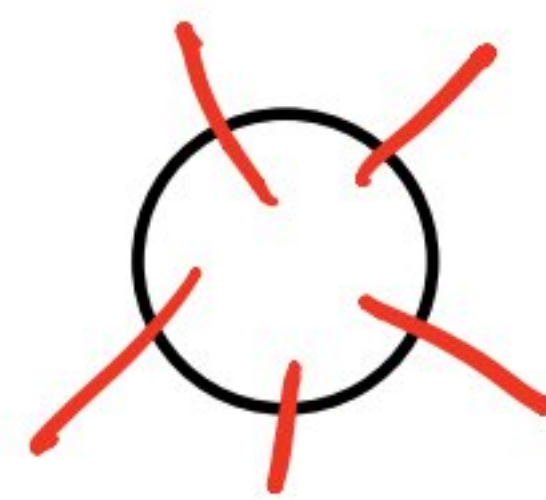
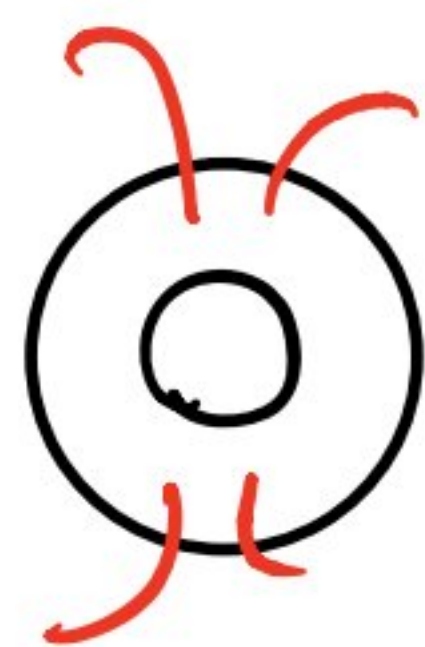
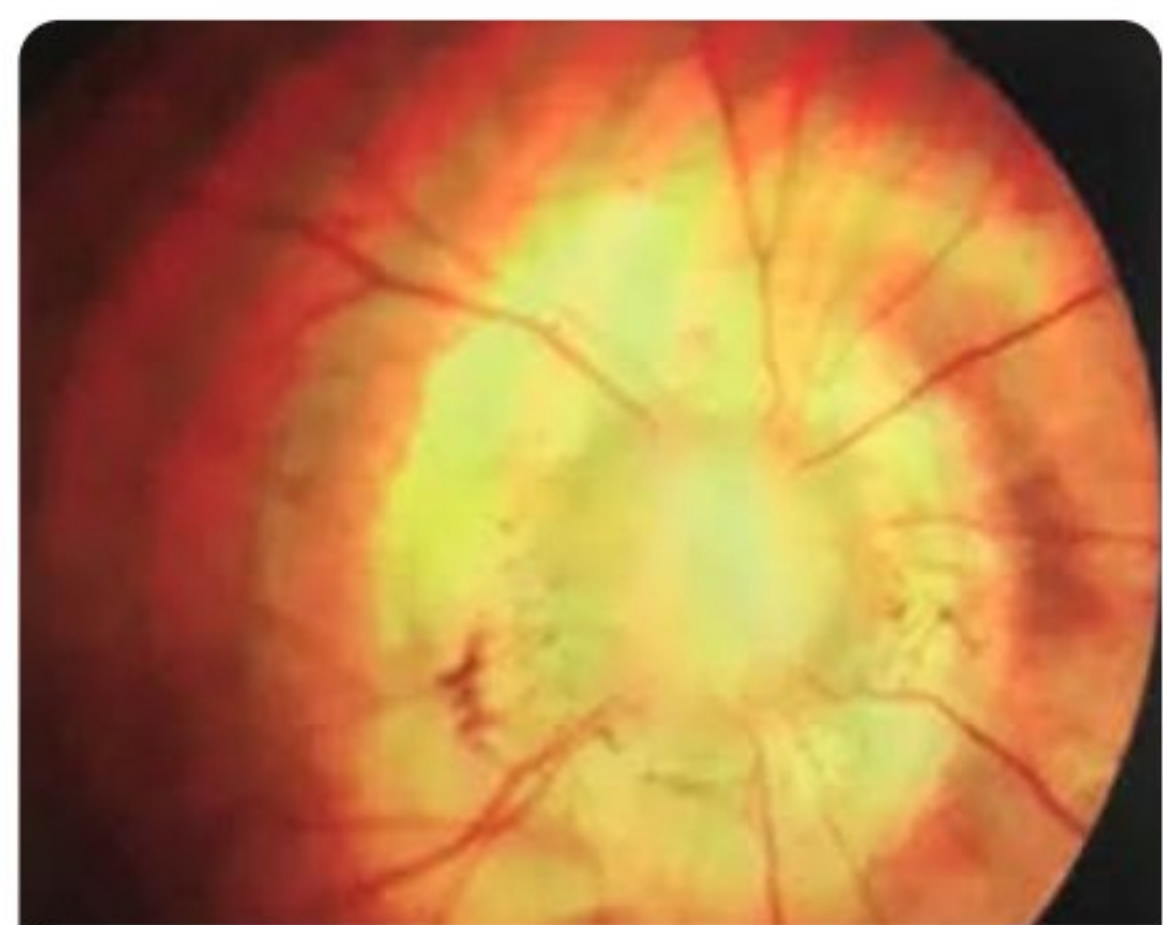
• Disc edema

• Fundus fluorescein angiography (FFA) → No dye leakage on disc
→ Normally dye leaks

• OA (IP)

• Sequential B/L ON involvement

• Congenital COLOBOMA of disc



Morning glory flower

OCULOMOTOR NERVE

NUCLEAR COMPLEX

Situated in midbrain @ level of superior colliculus

MOTOR Nucleus

2 paired subnuclei

CL SR

IL - MR
IR
IO

unpaired subnuclei

B/L LPS

EDINGER - WESTPHAL
preganglionic parasymp.
fibres

Sphincter pupillae
ciliary muscle

FASCICULAR

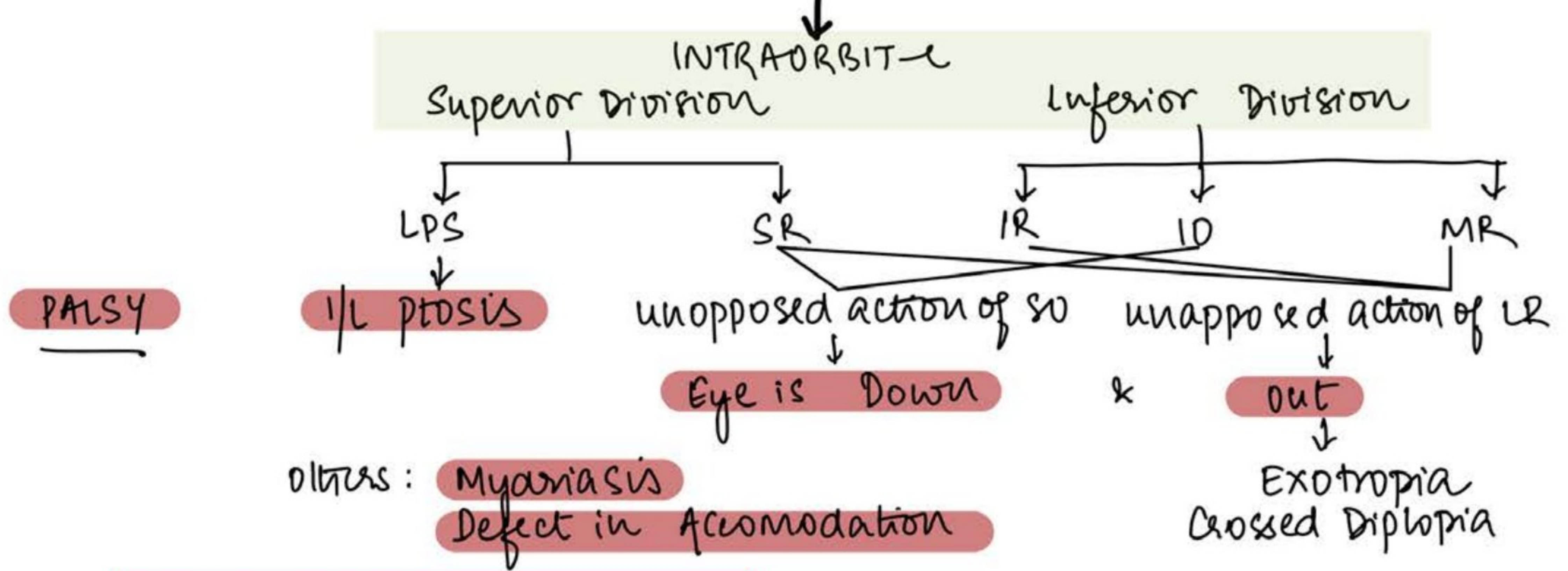
Fibers pass thro Red nucleus into interpeduncular fossa

BASILAR

Leaves midbrain on medial aspect of cerebral peduncle
passes b/w posterior cerebral (A) & superior cerebellar (A)

INTRACAVERNOUS

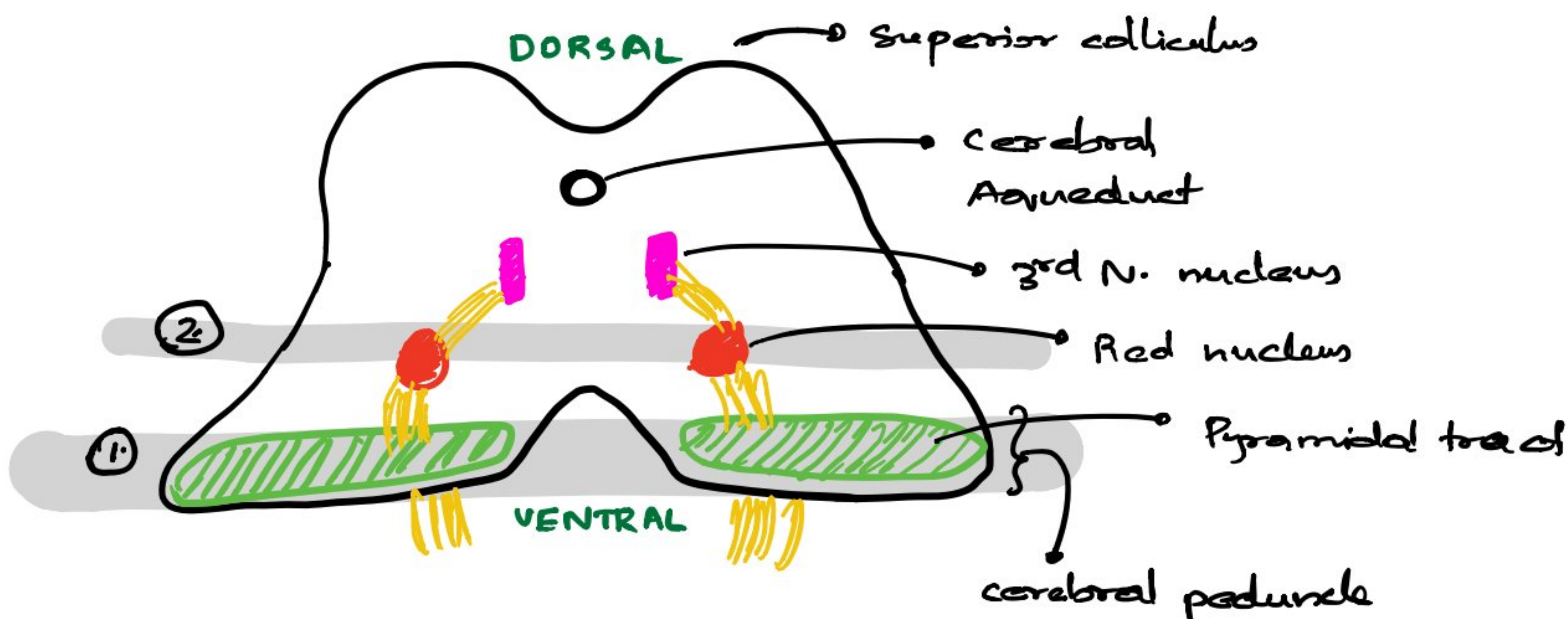
Runs in LATERAL wall above trochlear Nerve
Divides into Superior & inferior branches
Enters orbits thro annulus of Zinn



NUCLEAR PALSY	
B/L	ptosis
B/L	Myiasis & cycloplegia
C/L	SR weakness

Lesion @ 3rd Fascicles

- ① At cerebral pedicle → I/L 3rd N. Palsy
⊕
C/L Hemiplegia } **WEBER SYNDROME**



- ② Red nucleus → I/L 3rd N. Palsy
+ C/L Hemitremors } **BENEDICT SYNDROME**

"Red wine → Tremor → Addict"

- ③ Superior cerebellar peduncle } I/L TNP
+ I/L ataxia } **NOTHNAGEL SYNDROME**

- ④ **CLAUDE SYNDROME** } combination of Benedict + Nothnagel

①
②
③
④ } P. PCA infarct

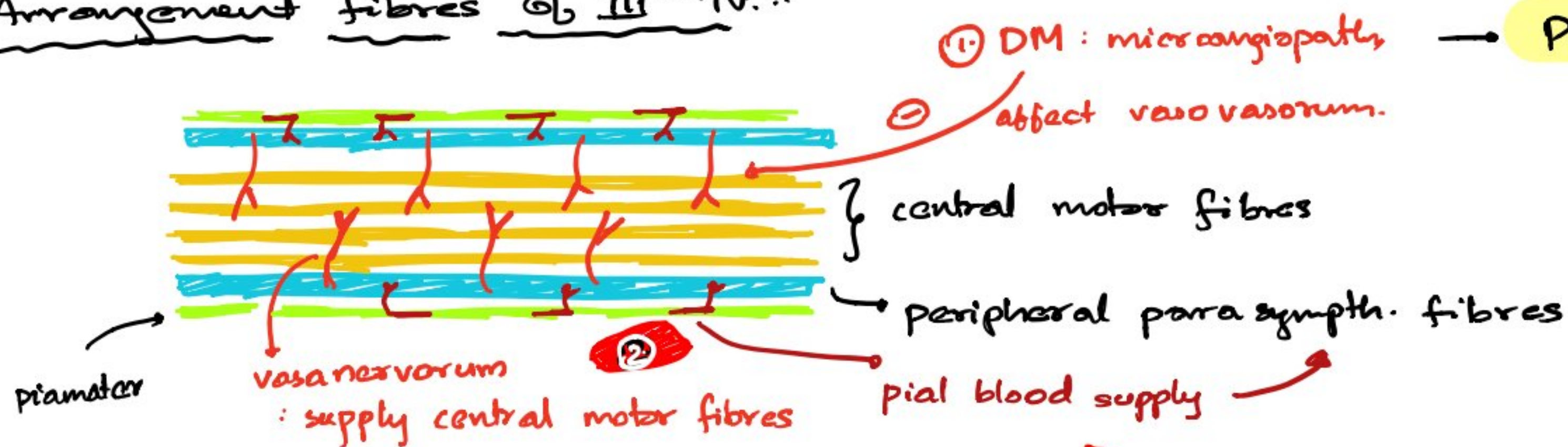


post. communicating Artery → Aneurysm → IIIrd N. palsy

Posterior cerebral Ar (above)

superior cerebellar Ar (below)

Arrangement fibres of IIIrd N.:



① DM: microangiopathy → Pupil sparing 3rd N. palsy

② affect vasa vasorum.

② Aneurysm →

Pial blood supply first affected

↓
pupillary dilatation

C/F of 3rd N. Palsy: "4D"

- 1) Drooping of lids
- 2) Deviation of eyeball → Down & out
- 3) Diplopia
- 4) Dilatation of Pupil → Neuroimage is mandatory to r/o posterior communicating artery aneurysm

'3D' → DM → Medical TNP → spontaneous resolve
/ pupil sparing TNP

Q: 56yr old DM ♀, with vertical & horizontal diplopia, deviation of R eye, light reflex (N) → Direct, Passive movement of eyeball is (+), Active movement Absent in certain direction. What is it?

- a) 3rd N. palsy
- b) 6th N. palsy × → only horizontal diplopia
- c) 4th N. palsy
- d) 7th N. palsy ×

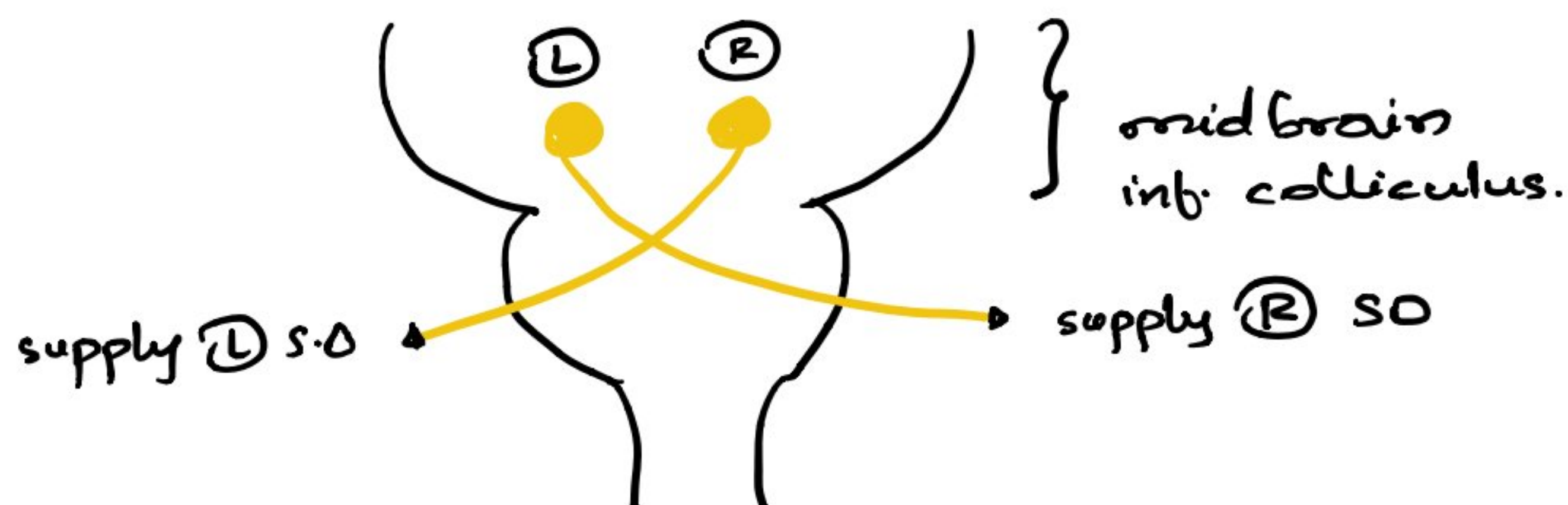
sol. Most likely N. to paralyse in DM is 6th N. > 3rd N

Q: Which statement regarding the innervation of the extraocular muscles is false?

- a) The levator palpebrae is innervated by fused central nucleus
- b) The superior oblique is innervated by ipsilateral IV nucleus.
- c) The inferior oblique is innervated by the ipsilateral III nucleus
- d) The superior rectus is innervated by the contralateral III nucleus [32%]

TROCHLEAR N. → Unique IVth N.

- Pure motor N.
- The only N. existing dorsally
- Supply **SO**



LONGEST INTRADURAL COURSE

- Thinnest cranial N.

Abducent N. {VIth N}

- Nucleus : PONS
- Supply: LR
- **Longest Subarachnoid Course**
- first N. paralysed in raised ICT

↳ passes over petroclinoid ligament

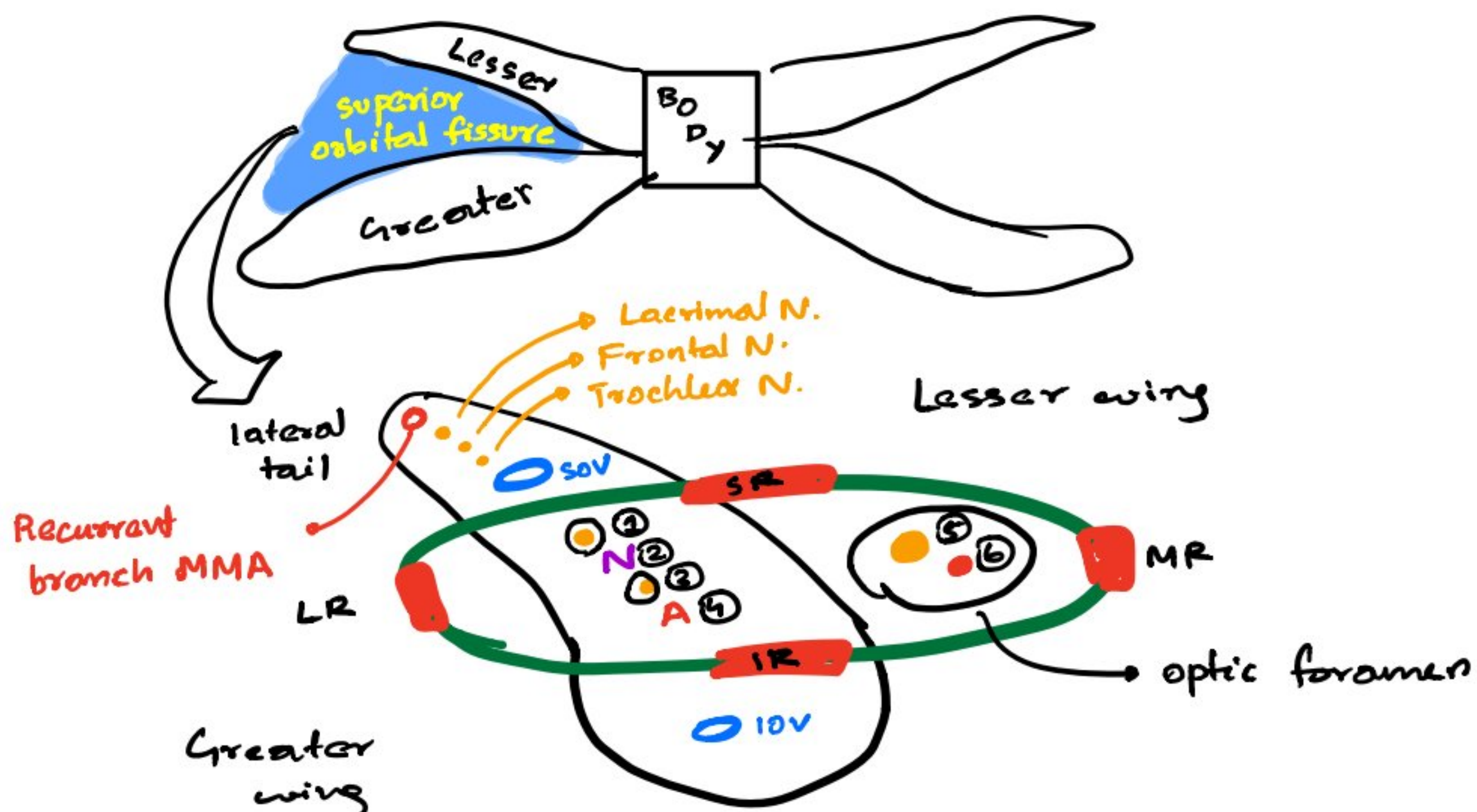
False localising sign

The first muscle to paralyse in raised ICT is lateral rectus.

↓
Abduction affected



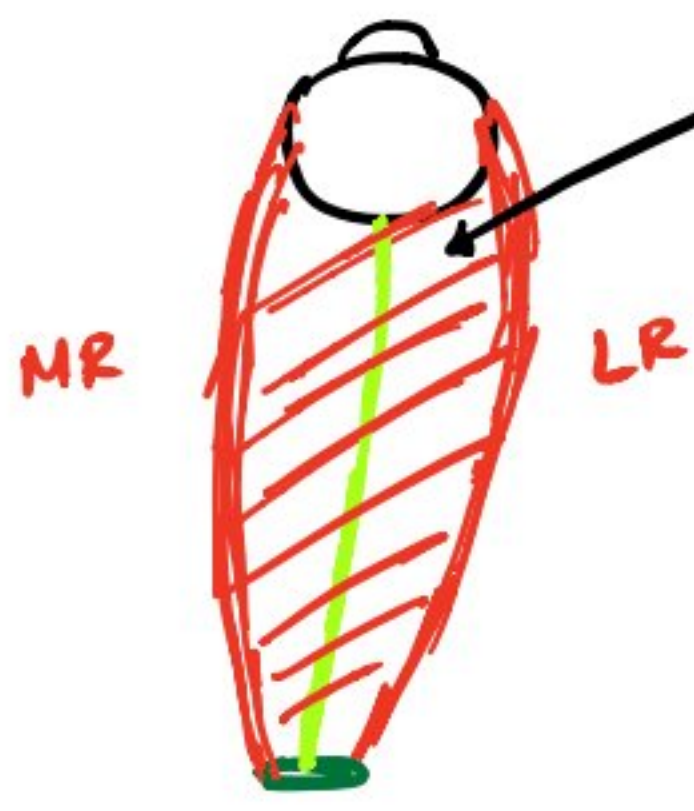
Sphenoid ::



- ① Oculomotor N. [superior div.]
- ② Nasociliary N.
- ③ Oculomotor N. [inferior div.]
- ④ Abducent N.
- ⑤ Optic N.
- ⑥ Ophthalmic Artery.

Inflammation in Apex of Orbit

- 1, Ophthalmoplegia
- 2, Corneal sensation affected



Retrobulbar Block

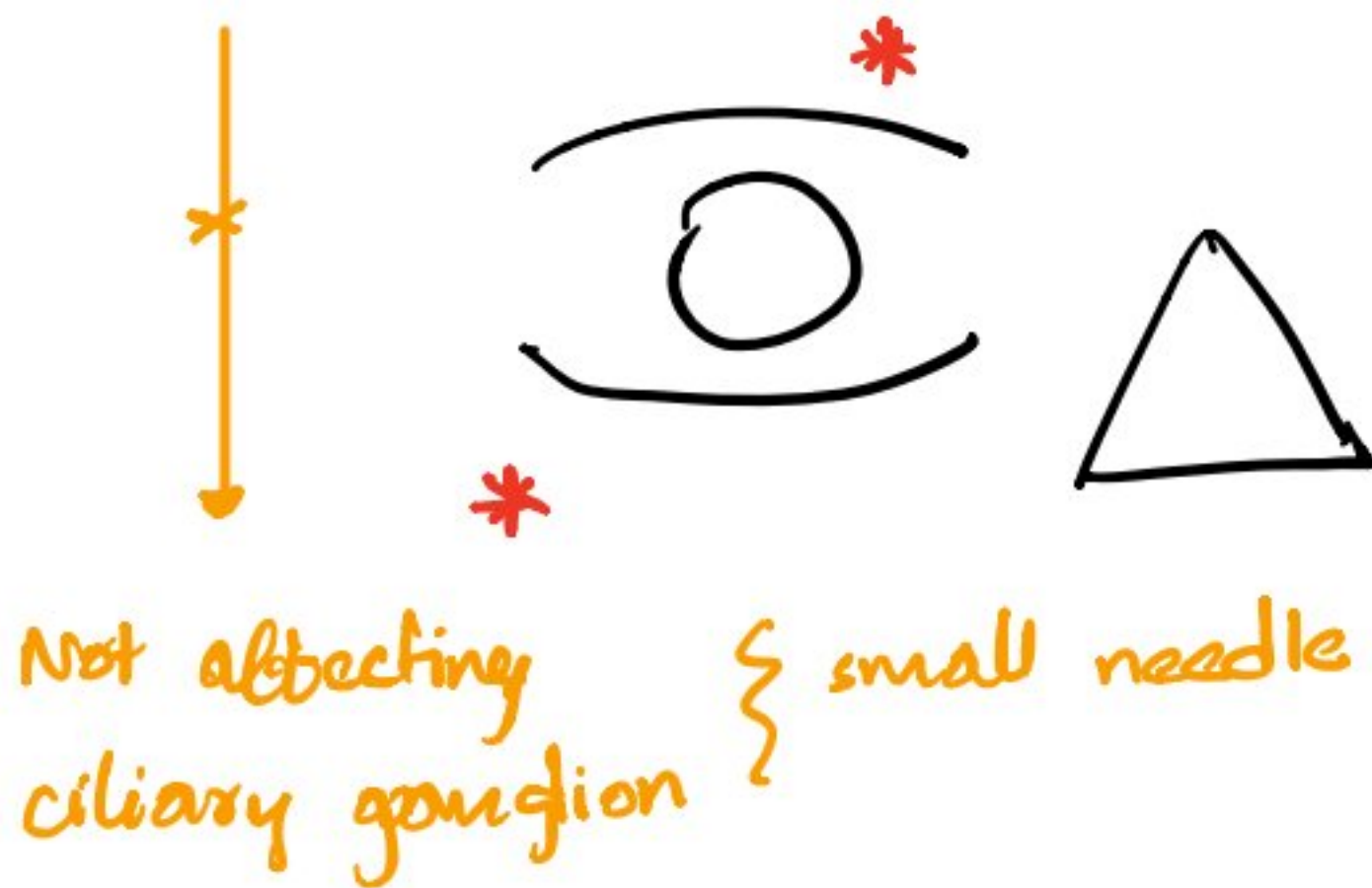
⇒ 2nd Trochlear N. last to be anesthetised (outside ring of Zinn)
Superior oblique last muscle to be paralysed

⇒ 2nd 10 → 1st muscle to be anesthetised

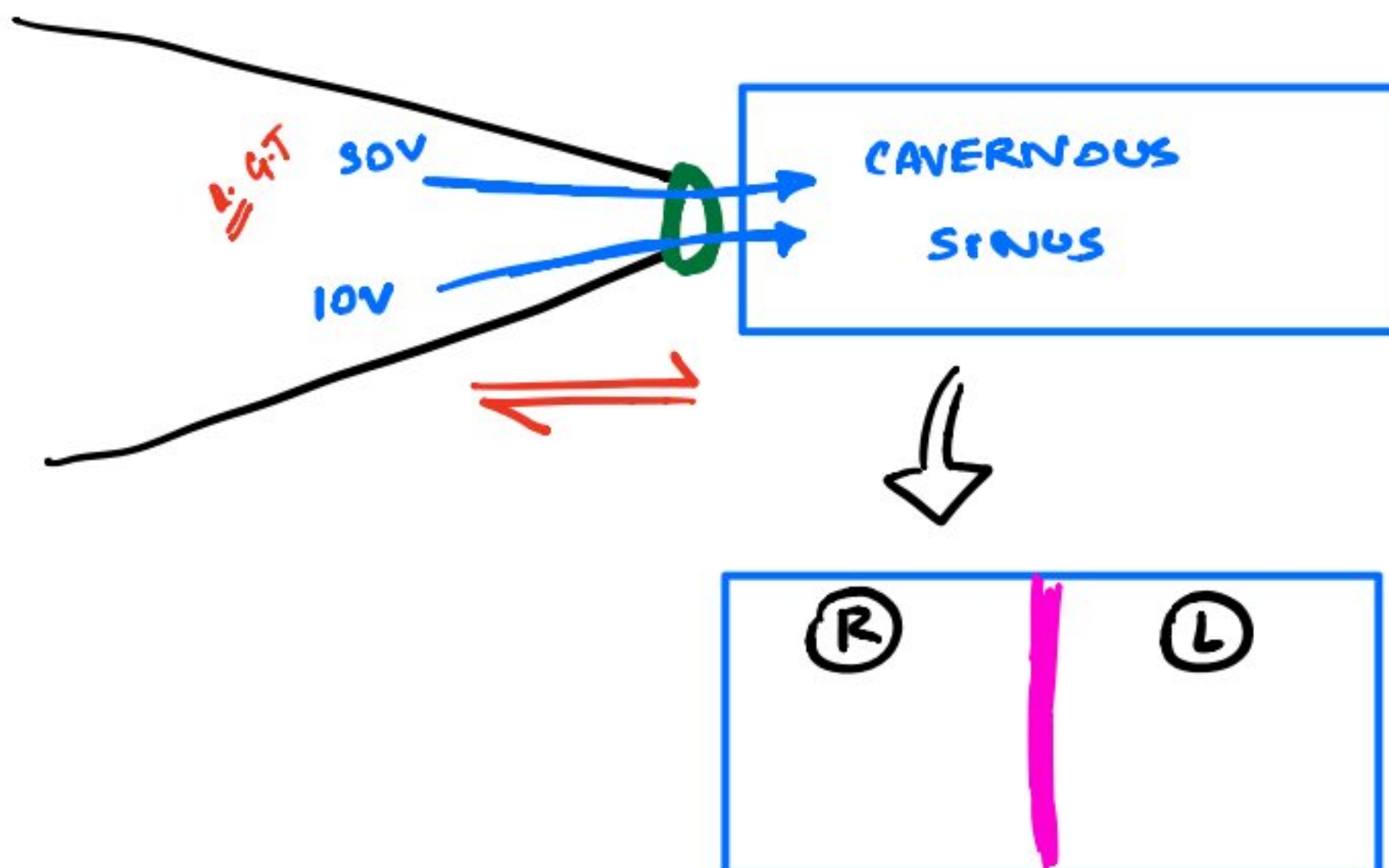
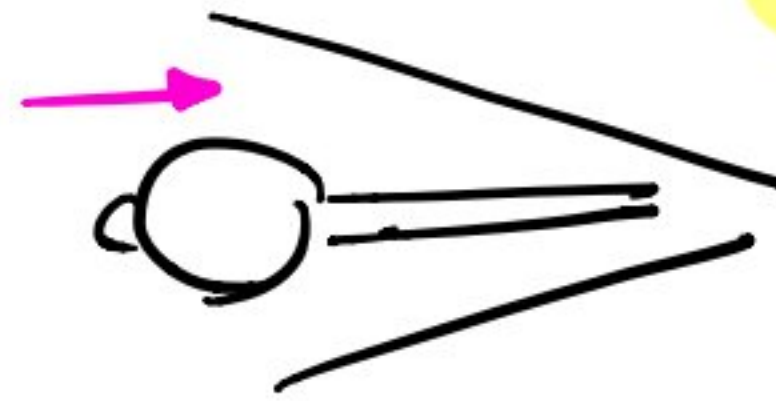
↳ ciliary ganglion = Pupillary dilatation paralysed

Peribulbar Block

first N. to be blocked = Trochlear N. (outside ring of Zinn)
= S.O



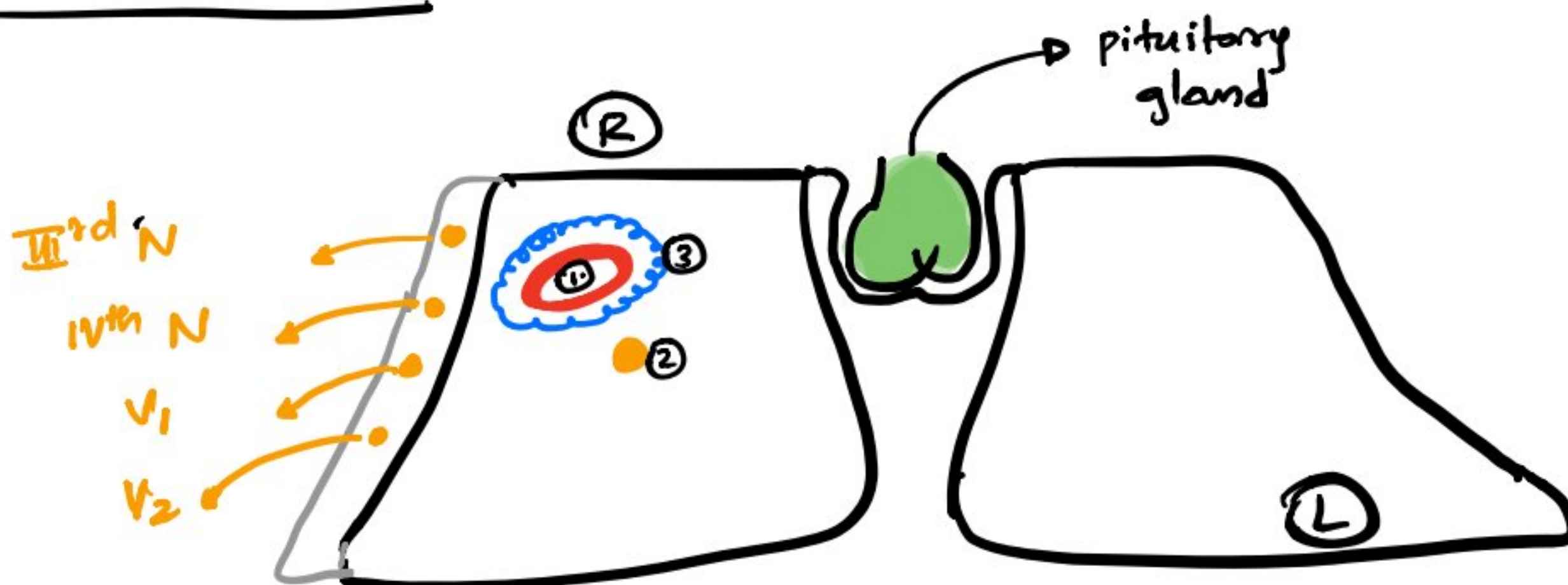
Not affecting ciliary ganglion { small needle



septum: incomplete posteriorly
R & L vein blood mixing.

• SOV & IOV are valveless → blood can flow to and from cavernous sinus.

CAVERNOUS SINUS:



ICA 2 As

- ① Internal carotid Ar
- ② Abducent N.
- ③ Sympathetic plexus

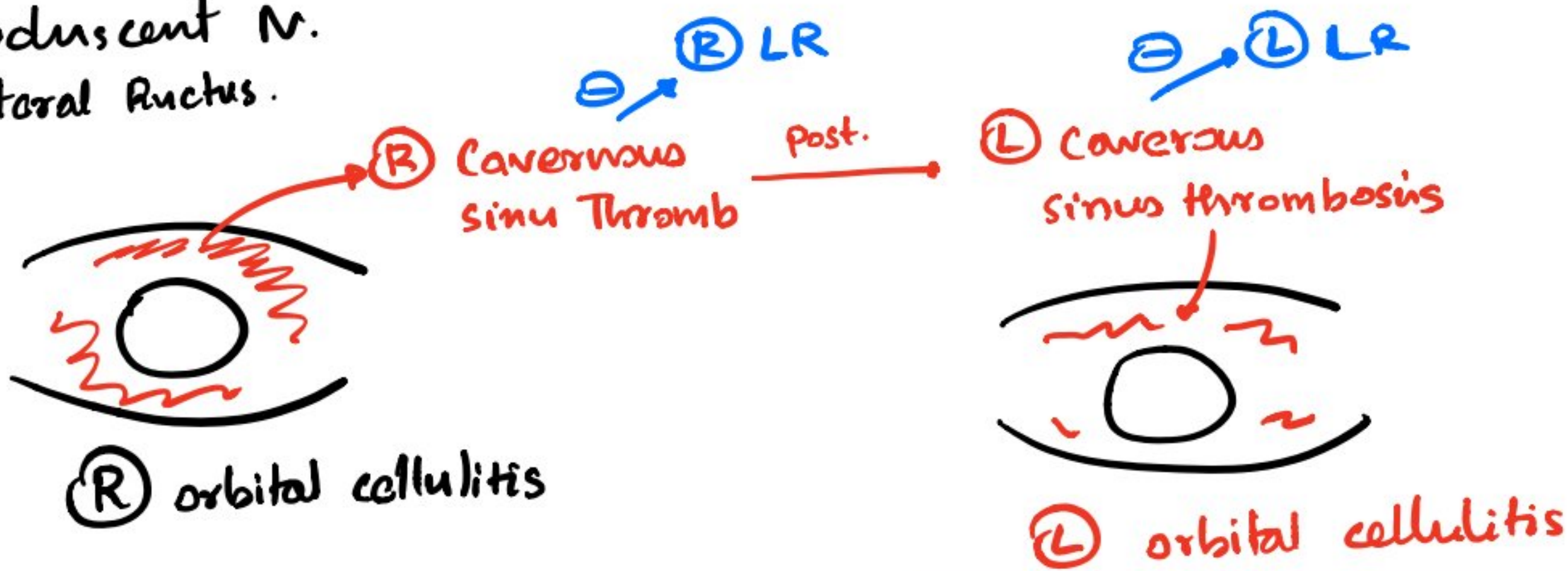
Q: Which structure absent in Cavernous Sinus?

a) V₁ ~~c) V₃~~

b) V₂ d) III

Q: 1st muscle to be paralysed in CST?

sol: Abducent IV.
Lateral Rectus.



TOLOSA HUNT Sx: Non specific granulomatous inflammation of Cavernous Sinus.

C/F 1) U/L Headache.

2) Ophthalmoplegia

3) Corneal sensation ↓ sed

Rx: Systemic steroids
Oral Prednisolone.

Q: Amaurosis fugax is characterized by all except?

a) Sudden onset

b) Transient nature

c) Monocular blindness

~~d) Irreversible nature~~

Pupillary Ab(N):

1) AMAUROTIC pupil → s/i complete Optical Atrophy
(Blind) (Glaucomatous)

• Fixed & dilated pupil

2) RAPD / Marcus Gunn Pupil → s/i optic neuritis U/L - (MC)

Also s/i :: i) Extensive RD

ii) CRAO

iii) Ischemic CRVO

iv) AION

Q: RAPD s/i in a/e:

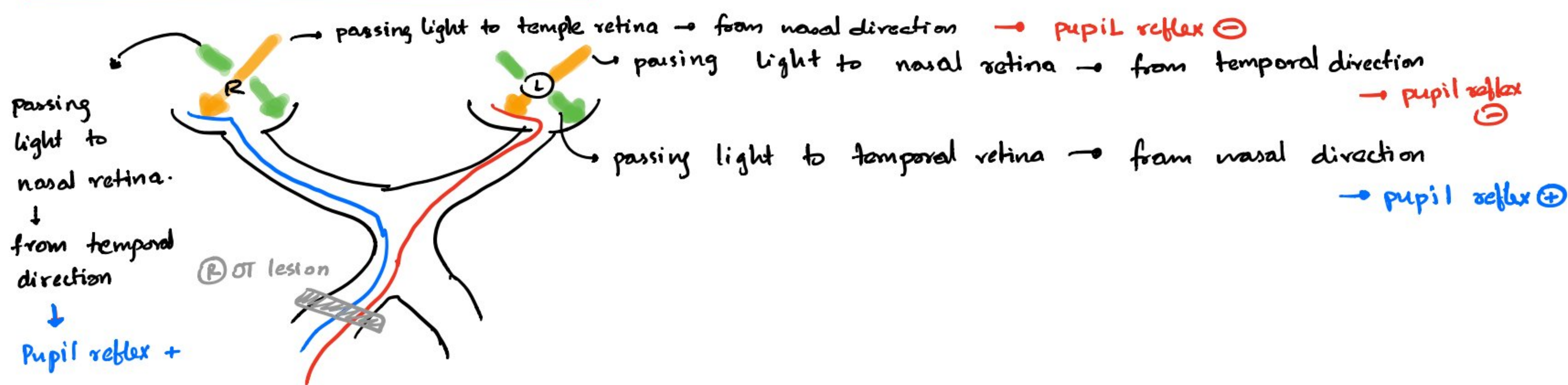
~~a) B/L ONitis~~

b) CRAO

c) CRVO

d) RD

WERNICKE'S HEMIANOPIC PUPIL → lesion of Distal optic tract



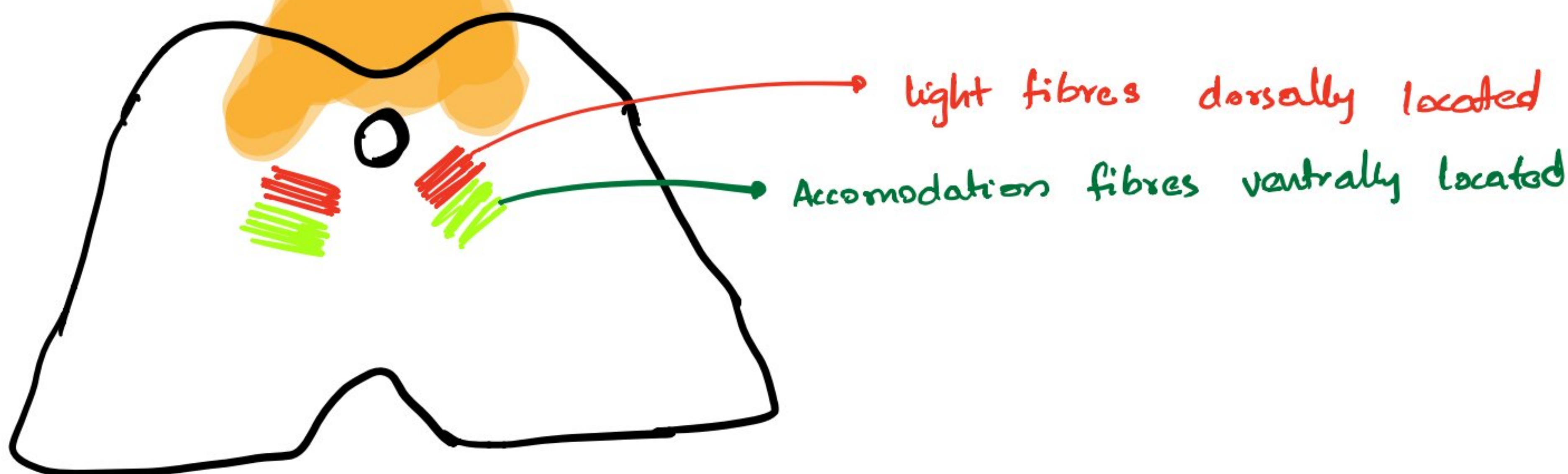
↳ ARGYLL-ROBERTSON Pupil (ARP) → light - near dissociation

- Light reflex absent
- accommodation reflex or near reflex present
- constricted pupil

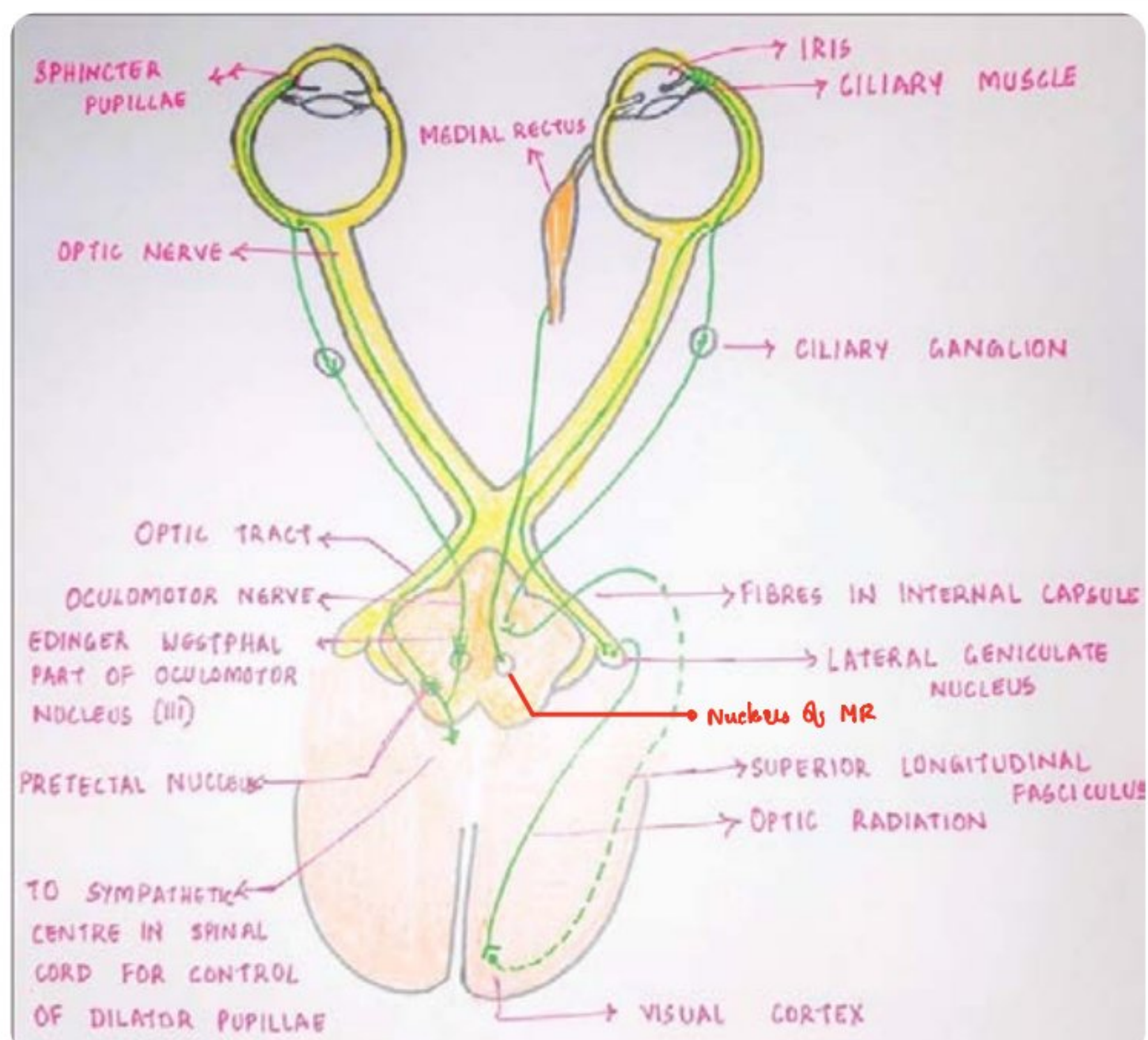
• Classical cause → **TABES DORSALIS & Neurosyphilis**

• (MC) cause → **Diabetes Mellitus**
→ **Alcohol**

TABES DORSALIS: 1st fibres affected light fibres



NEAR REFLEX / ACCOMMODATION REFLEX :: [B/L Phenomenon]



- 1) B/L pupillary constriction
- 2) Convergence
- 3) Accommodation